



VIRCL2 PROJECT DOCUMENTATION

Tristan De La Borda & Refik Mahic



[DATE]

LYCEE GUILAUME KROLL

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First weekly report

1. Start of the project

Within this section we will outline the start of the project which was during the autwo classes where we reviewed all we had to do and distributed the tasks. Within the first week the tasks were rather simple.

1.2 Installation of hypervisor

The hypervisor we were told to use is proxmox so we installed that on the servers given to us. We first used out laptops to make a bootable flash drive with the prxmox OS on it and when we plugged it into the server this graphical interface lead us though the installation process.



Figure 1:Proxmox instalation screen

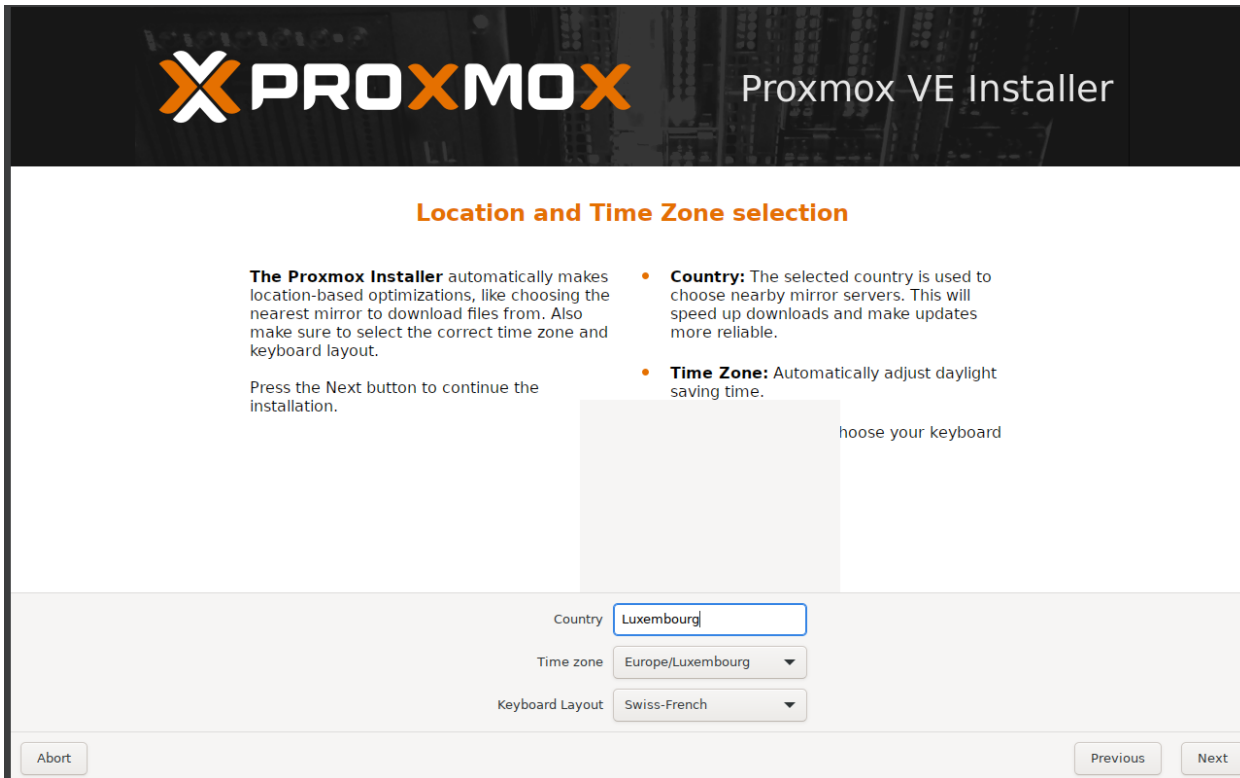
From here we selected the graphical user interface because of ease of use.

On the next section we were prompted to select the drive location for the Proxmox which we left at the USB what is there bellow in the table and with that afterwords proceeded.



Figure 2: Proxmox: Storage choosing

Next we selected the time zone and country which we also left at default or we input the country we are currently in. Also with the keyboard layout that we can choose for what we want



The Proxmox Installer

Location and Time Zone selection

The Proxmox Installer automatically makes location-based optimizations, like choosing the nearest mirror to download files from. Also make sure to select the correct time zone and keyboard layout.

Press the Next button to continue the installation.

- **Country:** The selected country is used to choose nearby mirror servers. This will speed up downloads and make updates more reliable.
- **Time Zone:** Automatically adjust daylight saving time.

Choose your keyboard

Country: Luxembourg

Time zone: Europe/Luxembourg

Keyboard Layout: Swiss-French

Abort Previous Next

Figure 3: Proxmox country and time

Then we were prompted to create a username and password for the administrator. We used our student emails for the emails of both servers and the password is “test..123” where we used to make the administrator



The screenshot shows the Proxmox VE Installer window. At the top, there is a header with the Proxmox logo and the text 'Proxmox VE Installer'. Below this, the title 'Administration Password and Email Address' is displayed in orange. The main content area contains two columns of text. The left column explains that Proxmox Virtual Environment is a full featured, highly secure GNU/Linux system based on Debian, and asks the user to provide the root password. The right column contains two bullet points: one for the password (must be at least 8 characters long, contain a combination of letters, numbers, and symbols) and one for the email (must be a valid email address for alerts). Below the text, there are three input fields: 'Password' (masked with dots), 'Confirm' (masked with dots), and 'Email' (containing 'mail@example.invalid'). At the bottom, there are three buttons: 'Abort', 'Previous', and 'Next'.

Proxmox Virtual Environment is a full featured, highly secure GNU/Linux system, based on Debian.

In this step, please provide the *root* password.

- **Password:** Please use a strong password. It must be at least 8 characters long, and contain a combination of letters, numbers, and symbols.
- **Email:** Enter a valid email address. Your Proxmox VE server will send important alert notifications to this email account (such as backup failures, high availability events, etc.).

Press the Next button to continue the installation.

Password:

Confirm:

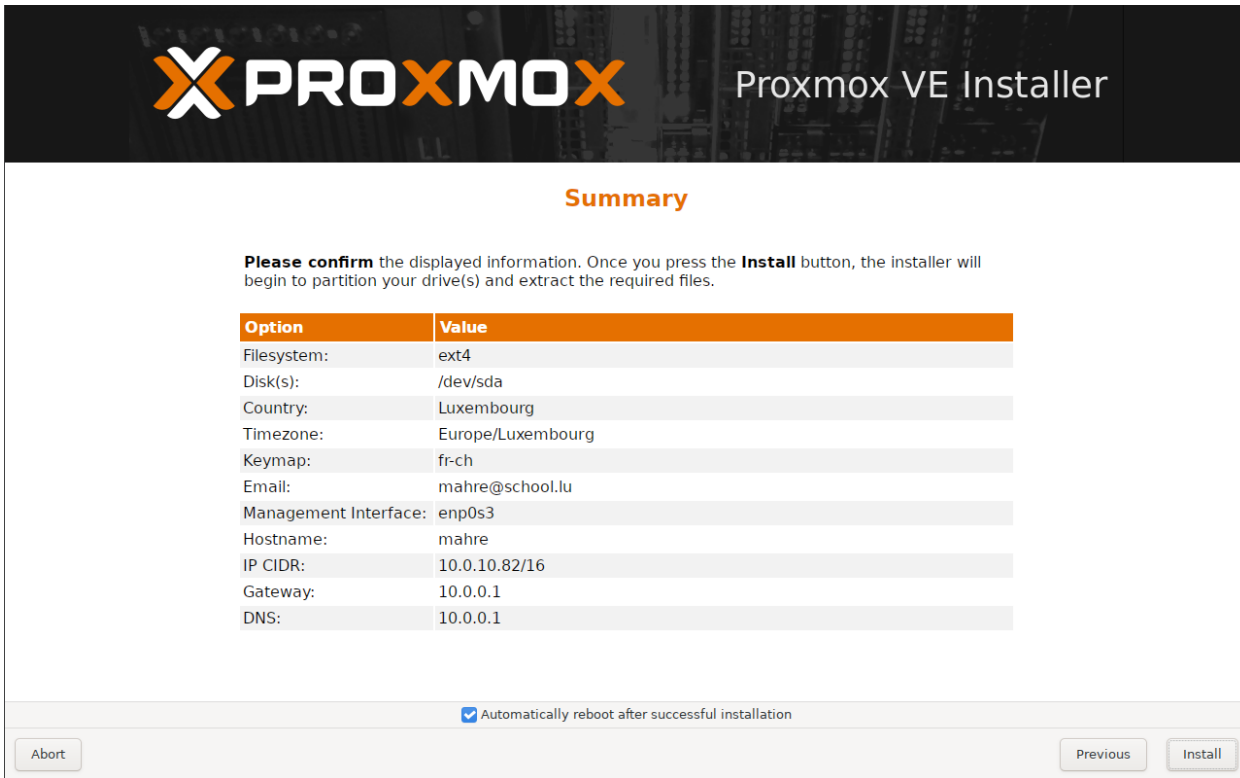
Email:

Figure 4: Proxmox password and email

Next we had to configure the network specifications for the servers network which we setup as following:

Using the provided server IP range given to us.

For both of the servers we used the ip address from the range



The image shows the Proxmox VE Installer Summary screen. At the top, the Proxmox logo and 'Proxmox VE Installer' text are displayed. Below this is a 'Summary' section with a warning: 'Please confirm the displayed information. Once you press the Install button, the installer will begin to partition your drive(s) and extract the required files.' A table lists various configuration options and their values. At the bottom, there is a checkbox for 'Automatically reboot after successful installation' which is checked, and three buttons: 'Abort', 'Previous', and 'Install'.

Option	Value
Filesystem:	ext4
Disk(s):	/dev/sda
Country:	Luxembourg
Timezone:	Europe/Luxembourg
Keymap:	fr-ch
Email:	mahre@school.lu
Management Interface:	enp0s3
Hostname:	mahre
IP CIDR:	10.0.10.82/16
Gateway:	10.0.0.1
DNS:	10.0.0.1

☒ Automatically reboot after successful installation

Abort Previous Install

Figure 5: Proxmox installation overview

This is the final step after the installation process begins and we just had to wait some time.



Figure 6: Final Proxmox installation screen

After the installation there we get the to the dashboard of proxmox with the Login of the user that we have created before

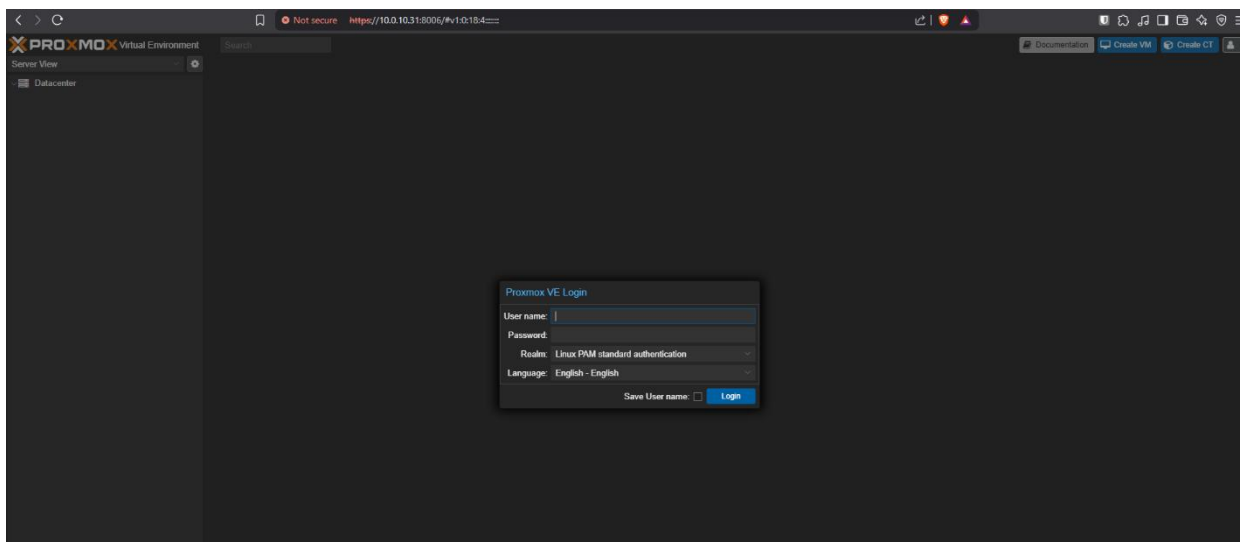


Figure 7: Proxmox Login screen

After we logged in the dashboard with our credentials we get this view is of the dashboard where on the left we have the hostname virtual machine and storage devices on the right we can see more information to the thinks we have on the right in this screen shot we can see the connection of two servers my and Tristans that we have put in a cluster

Type	Description	Disk usage...	Memory us...	CPU usage	Uptime	Host CPU ...	Host Mem...	Tags
node	delta	25.4 %	31.8 %	0.3% of 24 ...	22:53:18			
node	mahre	7.7 %	9.2 %	0.2% of 24 ...	1 day 01:57:26			
qemu	101 (VM 101)	0.0 %	87.0 %	0.4% of 10 ...	22:49:00	0.2% of 24 ...	22.2 %	
qemu	100 (VM 100)	0.0 %	17.2 %	0.2% of 4 ...	22:50:08	0.0% of 24 ...	2.2 %	
sdn	localnetwork (delta)							
sdn	localnetwork (mahre)							
storage	ProxmoxHDD (delta)	4.1 %						
storage	local (delta)	25.4 %						
storage	local-lvm (delta)	0.0 %						
storage	truenas (delta)	0.0 %						
storage	ProxmoxHDD (mahre)	0.7 %						
storage	local (mahre)	7.7 %						
storage	local-lvm (mahre)	0.0 %						
storage	truenas (mahre)	0.0 %						

Figure 8: Proxmox dashboard

Add a NAS

To Add a Nas we will need to do this steps:

The first step is to go to Datacenter and the to Storage. When you did that you will have a window like this

ID	Type	Content	Path/Target	Shared	Enabled	Bandwidth Limit
ProxmoxHDD	Directory	Backup, Disk image, ISO image, Conta...	/mnt/pve/ProxmoxHDD	No	Yes	
local	Directory	Backup, ISO image, Container template	/var/lib/vz	No	Yes	
local-lvm	LVM-Thin	Disk image, Container		No	Yes	
truenas	NFS	Disk image	/mnt/pve/truenas	Yes	Yes	

Figure 9:Proxmox Storage

After that we press on the add above and then we chose the nfs what is for the is for the Nas but also before we do that we need to give the permission on the nas for the ip that is form the server so that is allowed after we do that we will get this window. We write the id server and what else is needed and then we just press on add and that is how we add an nass to Proxmox

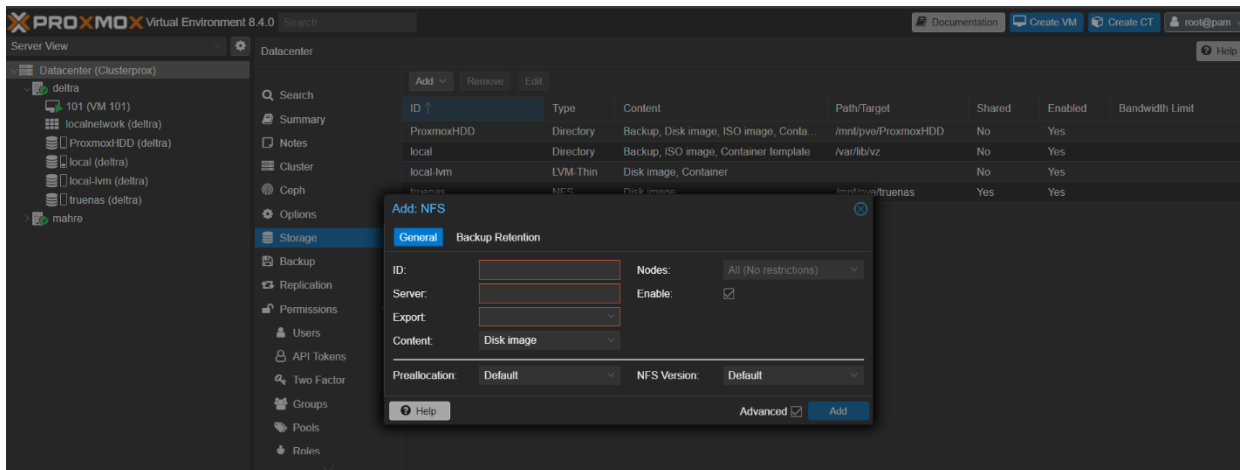


Figure 10:Proxmox add NFS

Creation of an VM

When creating a Virtual Machine (VM), there are a few important considerations to keep in mind. There are two primary methods for VM creation: using the Graphical User Interface (GUI) or the command-line console. In this guide, we will focus on how to create a VM using the GUI.

Step 1: Check System Resources

Before starting, ensure that your physical host system has sufficient computing resources. This includes enough CPU cores, RAM, and disk space to support the virtual machine. Without adequate resources, the VM may perform poorly or fail to start.

Step 2: Upload the ISO Image

Once you've confirmed that the host system is capable of running a VM, the next step is to upload the ISO file. This ISO file serves as the installation media for the operating system you want to install on the VM. You should upload the ISO to the appropriate directory or storage location used by your virtualization platform (for example, the "ISO Library" or a dedicated datastore, depending on the software you're using).

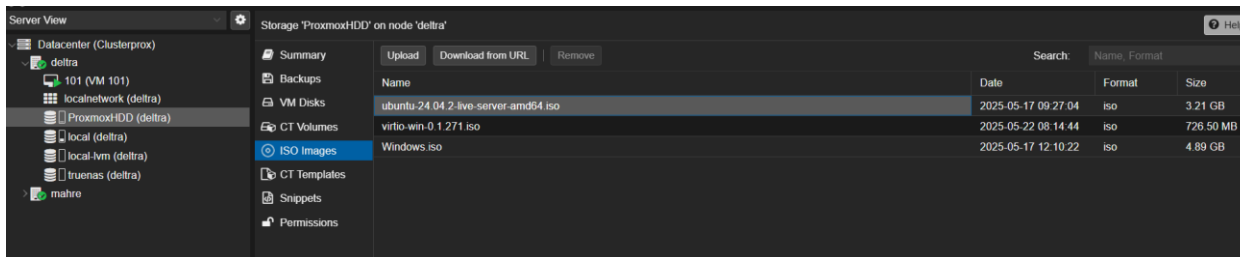


Figure 11: Proxmox iso tab

When we uploaded the iso to the server then we can press in the top right corner

Create VM

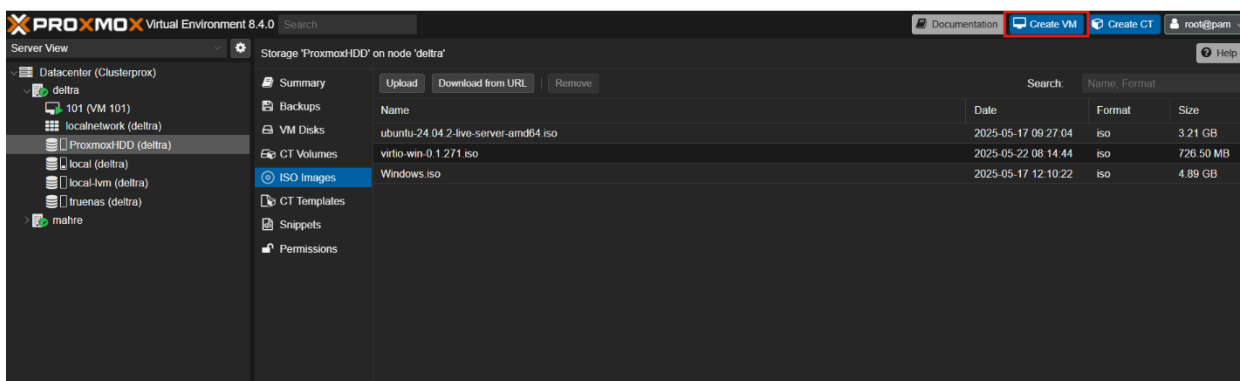


Figure 12: Proxmox creation of VM

After we pressed the button we will get this window to open

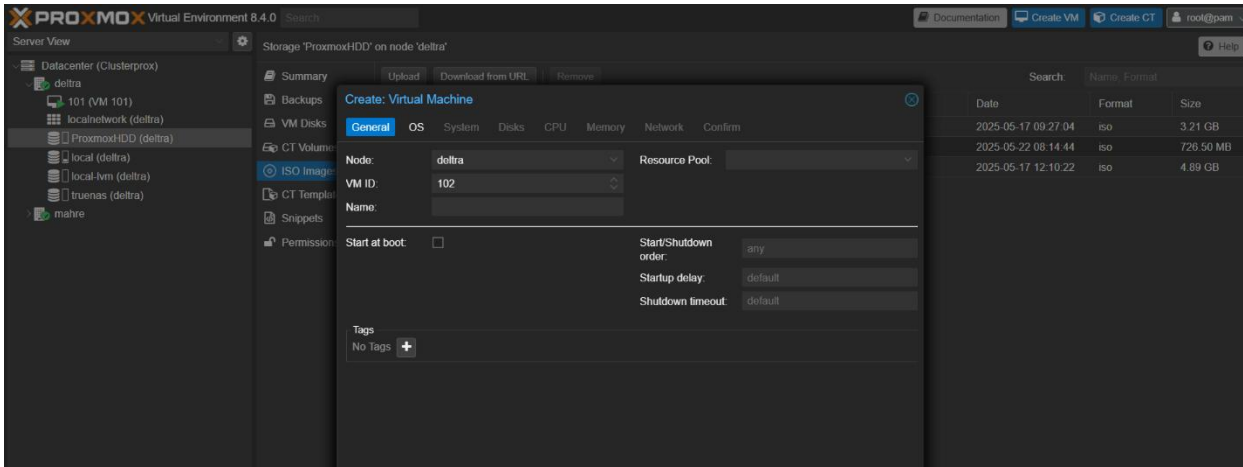


Figure 13: Proxmox Creation of an VM

Step 3: Configure the VM Settings

The Proxmox wizard includes multiple steps. Here's a breakdown of each: Select the Proxmox node where the VM will be created. A unique identifier will be generated automatically, or you can set your own. Enter a name for your VM that clearly describes its purpose (e.g., ubuntu-server-01). Select the ISO file you uploaded earlier from the dropdown. Guest: Choose Linux, Windows, etc., depending on your ISO. Select the closest matching version if available. Configure options like BIOS type (usually OVMF for UEFI or SeaBIOS for legacy), SCSI controller, and graphics card (like Default, Spice, or VirtIO). Bus/Device: Choose VirtIO for best performance. Pick the storage location (e.g., local-lvm). Set the disk size according to your needs (e.g., 20 GB for a small Linux server). Allocate the number of cores and optionally configure the CPU type (e.g., host for maximum compatibility/performance). Assign the amount of RAM. For example, 2048 MB (2 GB) for lightweight Linux servers. Select the network bridge (usually vmbr0) and model (e.g., VirtIO).

Step 4: Starting the VM and Installing the Operating System

Once the VM is created, it will appear in the left-hand pane under the node you selected. To begin the installation of the operating system:

1. Select the VM from the list.

2. In the main panel, click Start to power on the virtual machine.
3. Once the VM is running, go to the Console tab.

The Proxmox console provides a virtual display where you can interact with the VM as if you were sitting in front of a physical machine. If everything is configured correctly, the VM will boot from the ISO image you mounted earlier.

You should now see the installation screen of the operating system (such as Ubuntu, CentOS, Debian, Windows, etc.). Follow the on-screen instructions to complete the OS installation process.

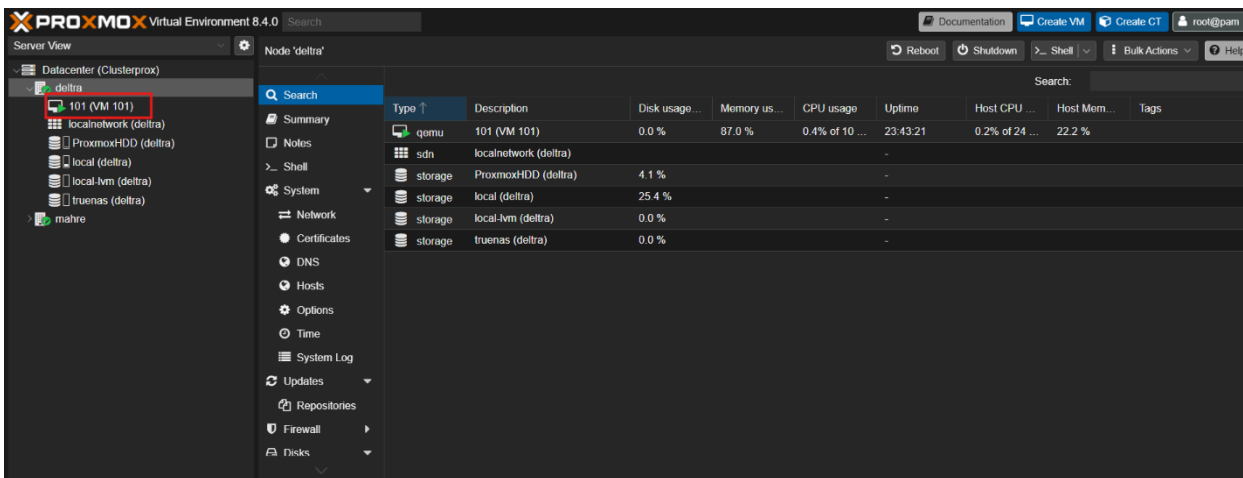


Figure 14: Proxmox VM installed

Step 5: Post-Installation Steps

After successfully installing the operating system inside the VM, there are a few recommended steps to complete the setup:

1. Remove the ISO Image (Optional but Recommended)

Once the OS is installed, go to the Hardware tab of the VM, select the CD/DVD drive, and remove or disconnect the ISO. This prevents the VM from booting into the installer again on the next reboot.

2. Install Guest Agents (QEMU Guest Agent)

Installing the QEMU Guest Agent enhances functionality between the host and VM. It allows for clean shutdowns, better reporting of IP addresses, and other advanced features. You can install it inside the VM (Linux or Windows) using the system's package manager or installer.

3. Configure Backups and Snapshots

Proxmox offers built-in backup and snapshot features. It's a good idea to set up regular backups, especially if the VM will host important applications or services.

4. Network Configuration

Ensure your VM has proper network access. You may need to configure static IP addresses, hostname, or DNS settings depending on your environment and use case.

At the end we have something like this

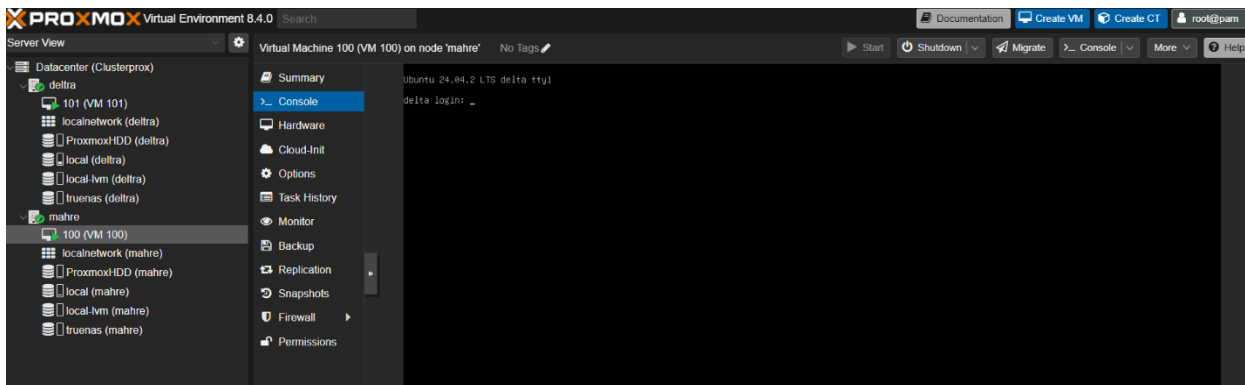


Figure 15:After installation

Migration Proxmox

Log in to the Proxmox Web Interface. In the left sidebar, select the VM you want to migrate. At the top of the main panel, click "Migrate". A pop-up window appears. Choose: Target Node: Select the destination node from the dropdown Migration Type: Leave default for online (if supported), Or choose to migrate with shutdown (offline). (Optional) Check the box for "with local disks" if you want to migrate VMs that use non-shared local storage (requires Proxmox to copy the entire disk). Click "Migrate" to begin the process.

Migrate VM 100 (VM 100)

Source node: mahre

Target node: deltra

Mode: Online

Target storage: Current layout

Info ↑

⚠ Migration with local disk might take long: ProxmoxHDD:100/vm-100-disk-0.qcow2 (50.00 GiB)

Help

Migrate

Figure 16: Proxmox Migration

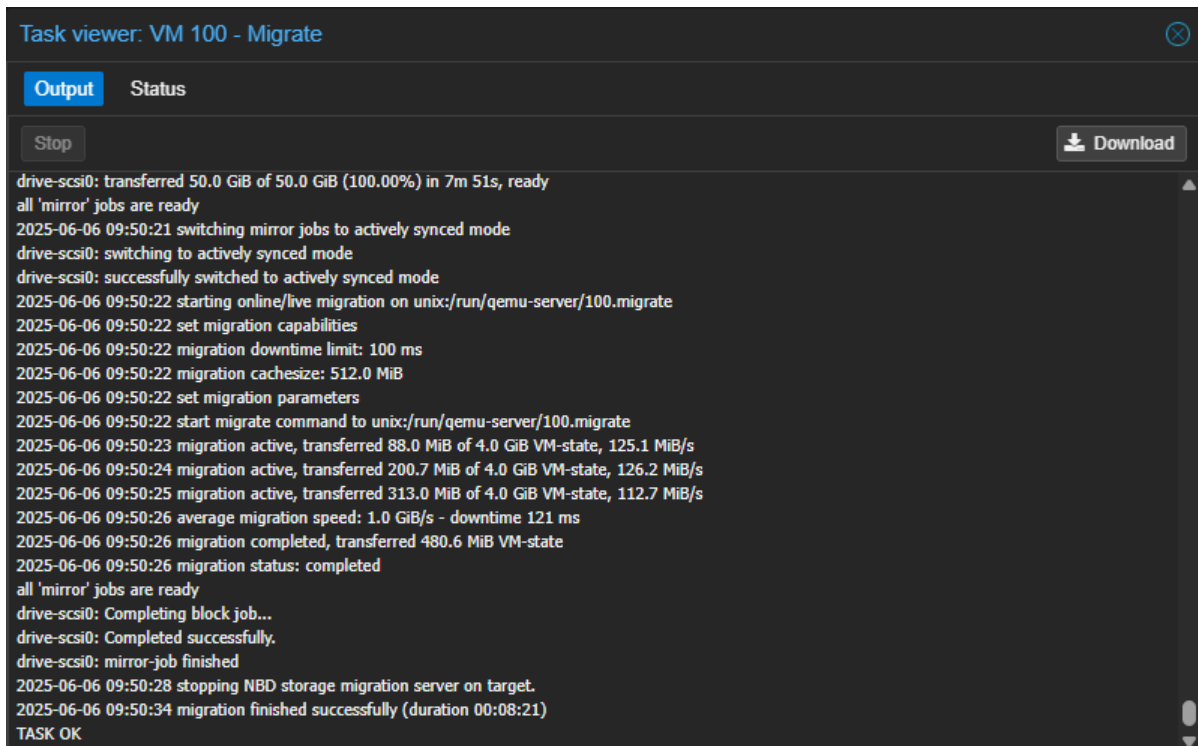
Task viewer: VM 100 - Migrate

Output Status

Stop Download

drive-scsi0: transferred 41.9 GiB of 50.0 GiB (83.84%) in 6m 35s
drive-scsi0: transferred 42.0 GiB of 50.0 GiB (84.06%) in 6m 36s
drive-scsi0: transferred 42.1 GiB of 50.0 GiB (84.28%) in 6m 37s
drive-scsi0: transferred 42.2 GiB of 50.0 GiB (84.50%) in 6m 38s
drive-scsi0: transferred 42.4 GiB of 50.0 GiB (84.72%) in 6m 39s
drive-scsi0: transferred 42.5 GiB of 50.0 GiB (84.93%) in 6m 40s
drive-scsi0: transferred 42.6 GiB of 50.0 GiB (85.15%) in 6m 41s
drive-scsi0: transferred 42.7 GiB of 50.0 GiB (85.37%) in 6m 42s
drive-scsi0: transferred 42.8 GiB of 50.0 GiB (85.59%) in 6m 43s
drive-scsi0: transferred 42.9 GiB of 50.0 GiB (85.81%) in 6m 44s
drive-scsi0: transferred 43.0 GiB of 50.0 GiB (86.03%) in 6m 45s
drive-scsi0: transferred 43.1 GiB of 50.0 GiB (86.25%) in 6m 46s
drive-scsi0: transferred 43.2 GiB of 50.0 GiB (86.47%) in 6m 47s
drive-scsi0: transferred 43.3 GiB of 50.0 GiB (86.69%) in 6m 48s
drive-scsi0: transferred 43.5 GiB of 50.0 GiB (86.91%) in 6m 49s
drive-scsi0: transferred 43.6 GiB of 50.0 GiB (87.13%) in 6m 50s
drive-scsi0: transferred 43.7 GiB of 50.0 GiB (87.35%) in 6m 51s
drive-scsi0: transferred 43.8 GiB of 50.0 GiB (87.59%) in 6m 52s
drive-scsi0: transferred 43.9 GiB of 50.0 GiB (87.81%) in 6m 53s
drive-scsi0: transferred 44.0 GiB of 50.0 GiB (88.03%) in 6m 54s
drive-scsi0: transferred 44.1 GiB of 50.0 GiB (88.25%) in 6m 55s
drive-scsi0: transferred 44.2 GiB of 50.0 GiB (88.47%) in 6m 56s
drive-scsi0: transferred 44.3 GiB of 50.0 GiB (88.69%) in 6m 57s
drive-scsi0: transferred 44.5 GiB of 50.0 GiB (88.91%) in 6m 58s

Figure 17: Migration status



```
Task viewer: VM 100 - Migrate

Output Status

Stop Download

drive-scsi0: transferred 50.0 GiB of 50.0 GiB (100.00%) in 7m 51s, ready
all 'mirror' jobs are ready
2025-06-06 09:50:21 switching mirror jobs to actively synced mode
drive-scsi0: switching to actively synced mode
drive-scsi0: successfully switched to actively synced mode
2025-06-06 09:50:22 starting online/live migration on unix:/run/qemu-server/100.migrate
2025-06-06 09:50:22 set migration capabilities
2025-06-06 09:50:22 migration downtime limit: 100 ms
2025-06-06 09:50:22 migration cachesize: 512.0 MiB
2025-06-06 09:50:22 set migration parameters
2025-06-06 09:50:22 start migrate command to unix:/run/qemu-server/100.migrate
2025-06-06 09:50:23 migration active, transferred 88.0 MiB of 4.0 GiB VM-state, 125.1 MiB/s
2025-06-06 09:50:24 migration active, transferred 200.7 MiB of 4.0 GiB VM-state, 126.2 MiB/s
2025-06-06 09:50:25 migration active, transferred 313.0 MiB of 4.0 GiB VM-state, 112.7 MiB/s
2025-06-06 09:50:26 average migration speed: 1.0 GiB/s - downtime 121 ms
2025-06-06 09:50:26 migration completed, transferred 480.6 MiB VM-state
2025-06-06 09:50:26 migration status: completed
all 'mirror' jobs are ready
drive-scsi0: Completing block job...
drive-scsi0: Completed successfully.
drive-scsi0: mirror-job finished
2025-06-06 09:50:28 stopping NBD storage migration server on target.
2025-06-06 09:50:34 migration finished successfully (duration 00:08:21)
TASK OK
```

Figure 18:End of migration

Cluster

Log in to the first Proxmox node using the web interface.

In the left sidebar, click on "Datacenter", then go to the "Cluster" tab.

Click the "Create Cluster" button at the top.

In the popup window:

Enter a Cluster Name (e.g., mycluster).

Choose the network interface for cluster communication (usually your main LAN IP).

Click "Create".

After the cluster is created, you'll see it listed in the Cluster tab.

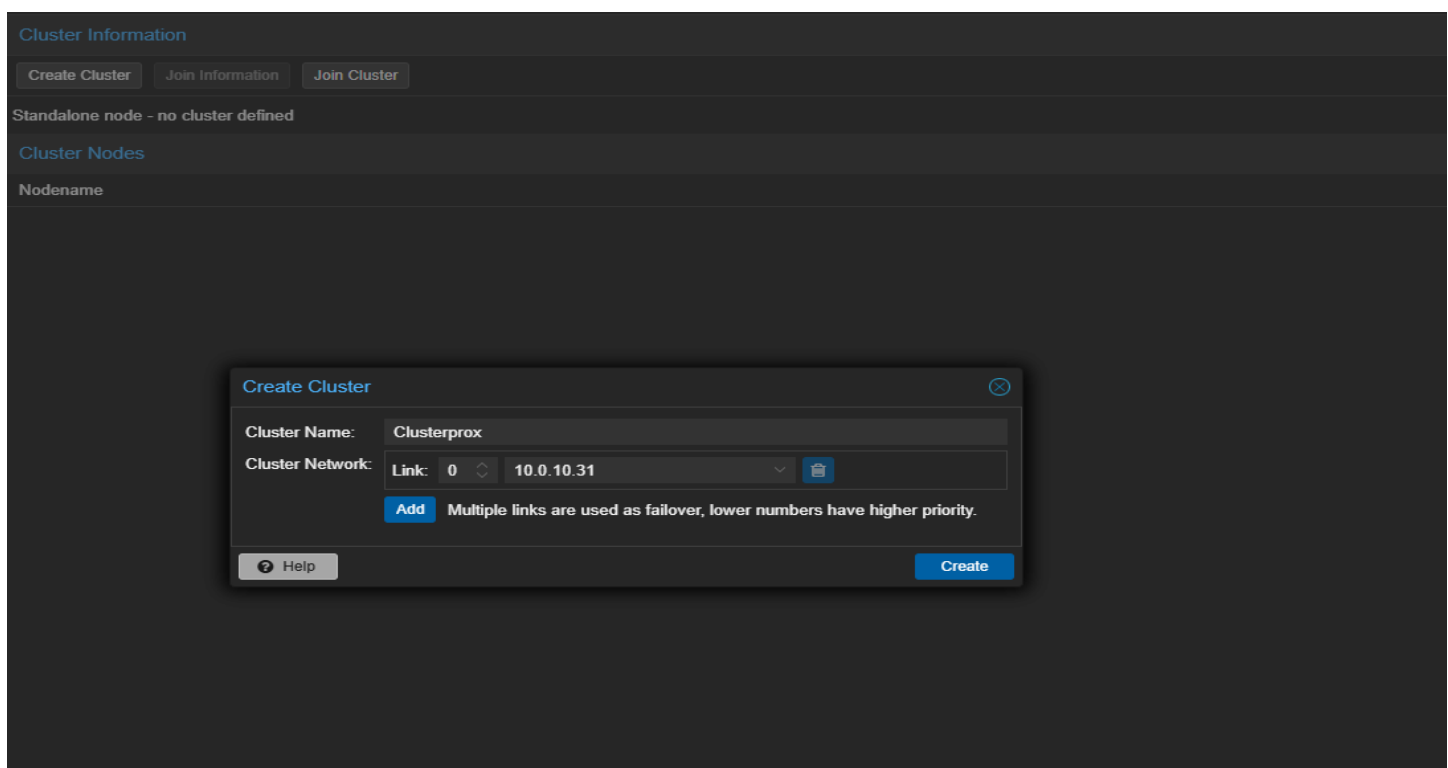


Figure 19:Cluster creation

Log in to the **second Proxmox node** (not yet in the cluster).

Go to **Datacenter > Cluster**, then click **“Join Cluster”**.

Fill in the form:

Enter the **IP address** of the first (master) node.

Enter the **root password** for that node.

Select the **network interface**.

Click **Join**. After a few moments, the node will be added to the cluster

Change RAM on a VM

Open the Proxmox Web Interface

Log in to your Proxmox VE Web UI.

In the left panel, click on the VM you want to modify.

2. Open the "Hardware" Tab

Click the “Hardware” tab from the top menu.

You’ll see the current hardware settings of the VM, including RAM.

3. Select the Memory Option

In the hardware list, click on "Memory".

A dialog will open showing the current allocated memory (RAM and optional ballooning settings).

4. Edit the RAM Value

Change the Memory (MB) field to the new amount of RAM you want to assign.

Optional: If you're using ballooning, adjust the Minimum Memory as well.

Note: If the VM is running, this action is a hotplug — and it will only work if:

The guest OS supports memory hotplug.

The VM was started with the hotplug option enabled (which is on by default in most cases).

5. Apply the Changes

Click “OK” to save the new RAM configuration.

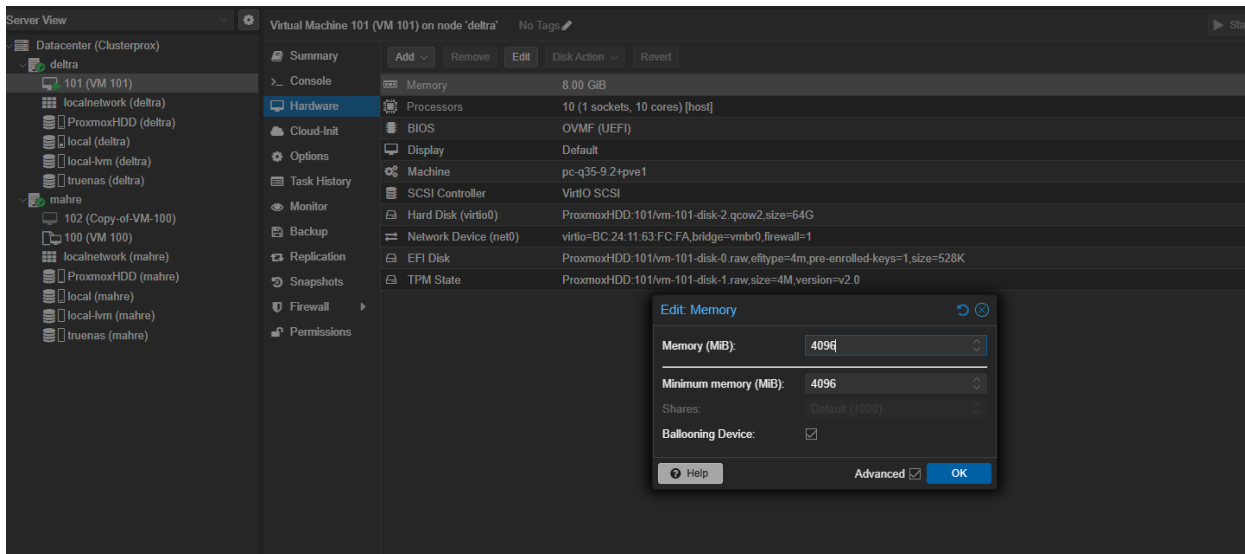


Figure 20: Edited memory live machine

Snapshots

Go to the "Snapshots" Tab

- Click the **"Snapshots"** tab from the top menu of the selected VM.

Click "Take Snapshot"

- Press the **"Take Snapshot"** button at the top.
- In the pop-up window:
 - **Name:** Give your snapshot a descriptive name (e.g., before-upgrade).
 - **Description:** (Optional) Add notes about why you're taking the snapshot.
 - **RAM:** Check the box if you want to include the current running memory (for full rollback of running state).

Click "Take Snapshot"

- Proxmox will freeze the VM briefly and create the snapshot.
- It will appear in the list with its name, date, and status.

How to Restore a Snapshot Select the Snapshot

- In the **Snapshots** tab, click on the snapshot you want to restore.

Click "Rollback"

- A warning will appear — restoring a snapshot will **replace the current state** of the VM with the snapshot.
- Click **"Yes"** to confirm.

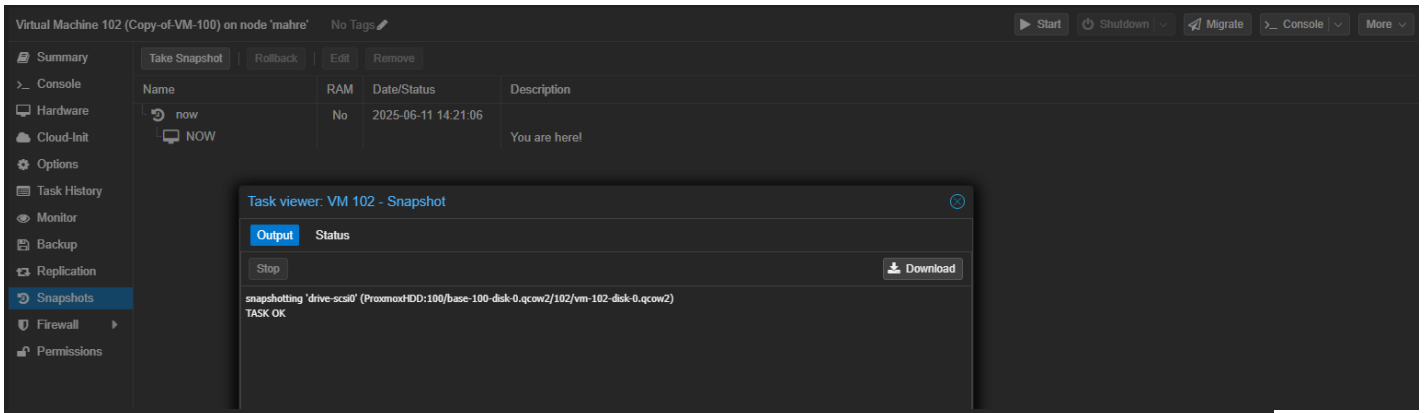


Figure 21: Creation of a snapshot

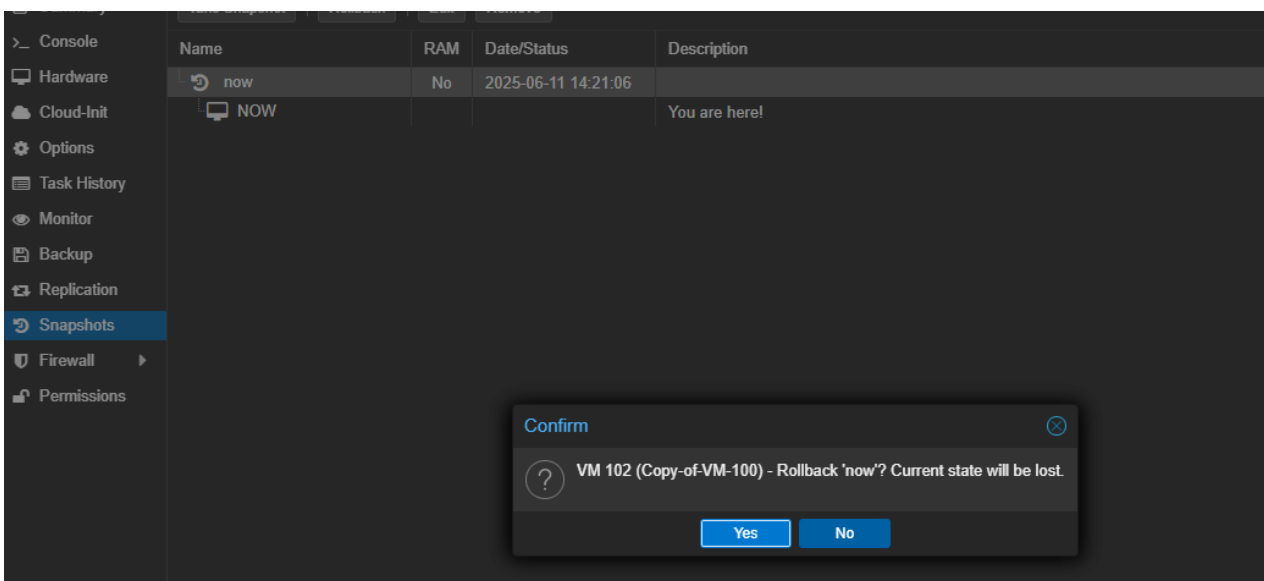


Figure 22: Created Snapshot

Template

Create and Configure a VM

- Set up a new VM as usual:
 - Install the operating system.
 - Install updates and any software/tools you want in your base image.
 - Clean up the system (optional: remove SSH keys, history, etc., for clean cloning).
 - **Do NOT start any services that shouldn't run on clones.**

Shut Down the VM

- The VM must be **stopped** before converting it to a template.
- Right-click the VM → **Shutdown**, or use the "**Shutdown**" button.

Convert to Template

- Right-click the VM in the left sidebar.
- Select "**Convert to Template**".
- Proxmox will turn the VM into a template — its icon will change to a square with a “T” inside.

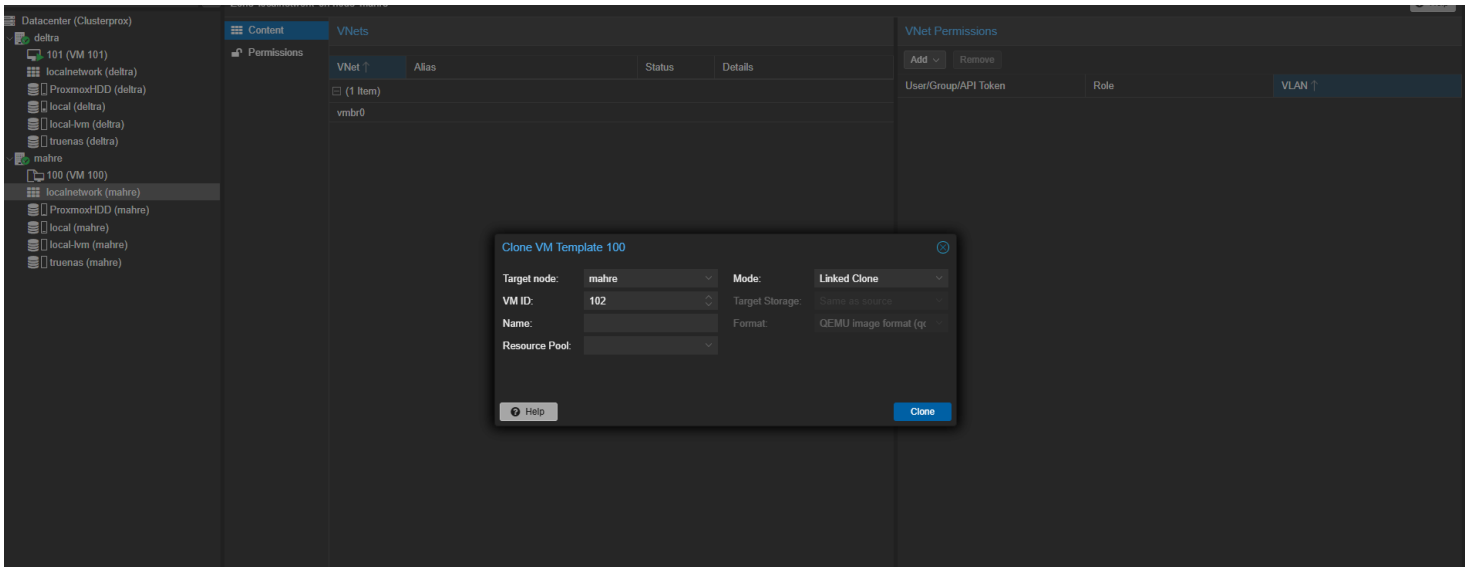


Figure 23: Creation of Template

Clone

Right-click the **VM or Template** in the left panel.

Click "**Clone**".

In the window that opens:

- Choose a **VM ID** and **Name**
- Select **Full** or **Linked** clone
- Choose the **target node** and **storage**

Click "**Clone**" to create it.

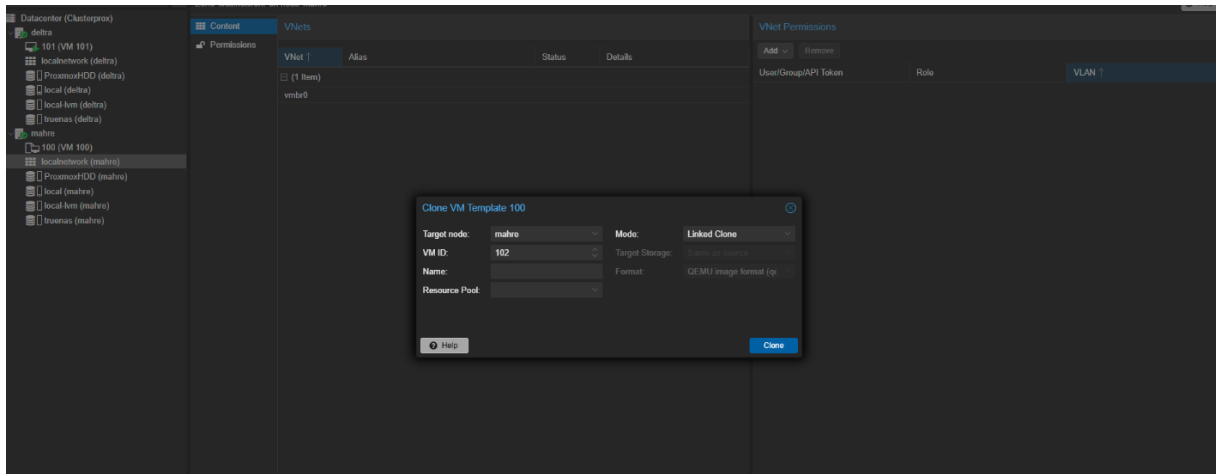


Figure 24:Cloning of a Template

Installation of XCP-ng

Step 1. Requirements

For the beginning of this installation, we will need some items.

- Installation media: XCP-ng Iso which can be acquired from their website at the following link: [XCP-ng](#)
- 50gb or more of free storage space on the hardware
- An internet connection

Step 2. Booting from the ISO

Once you have downloaded the ISO put the ISO on a bootable thumb stick and insert it into the server USB port. Once you enter your bios make sure to select the Thumb stick as the primary boot on the boot manager. Then save and restart.

For us it was installed on a Western Digital HDD:

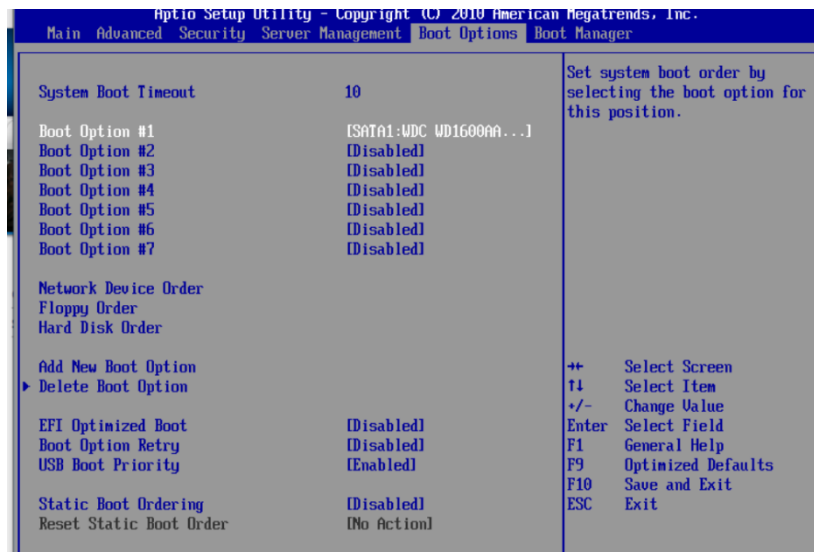


Figure 25:XCP BIOS boot

Step 3. Installation phase

Once XCP-ng ISO has launched you will be greeted with this screen.



Figure 26:XCP Welcome screen

From here you press enter to enter for the upgrade.

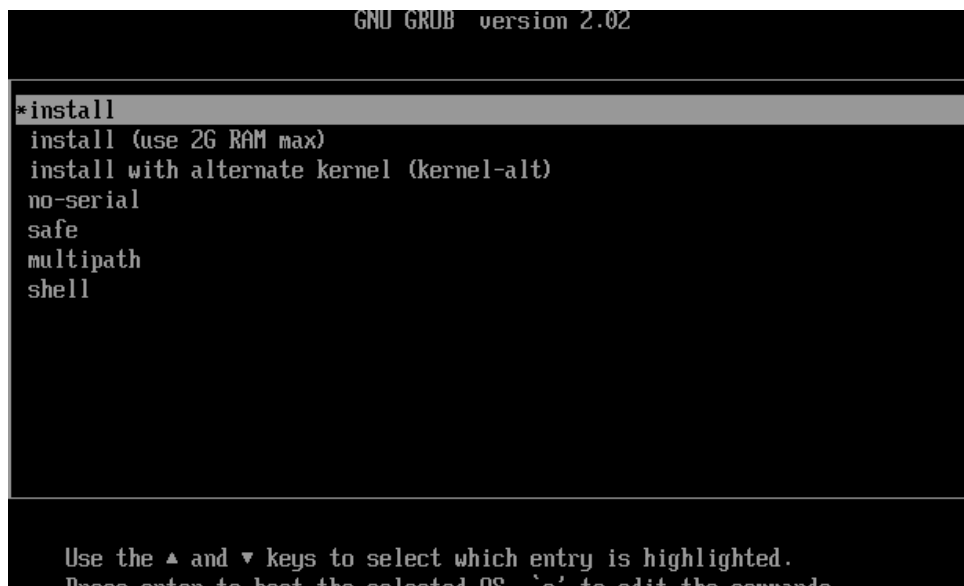


Figure 27:XCP Instalation screen

Then from the next screen select install.

The next few sections will be:

- a. Selection of keyboard: Select your prefer keyboard layout.
- b. Agreement to EULA: Agreement is mandatory to progress.
- c. Selection of primary installation drive: This will be where XCP is installed, select your desired location.
- d. Then selection of VM drive: This will be where the drives for the VMs are stored also select your desired location.

Next will be the installation media

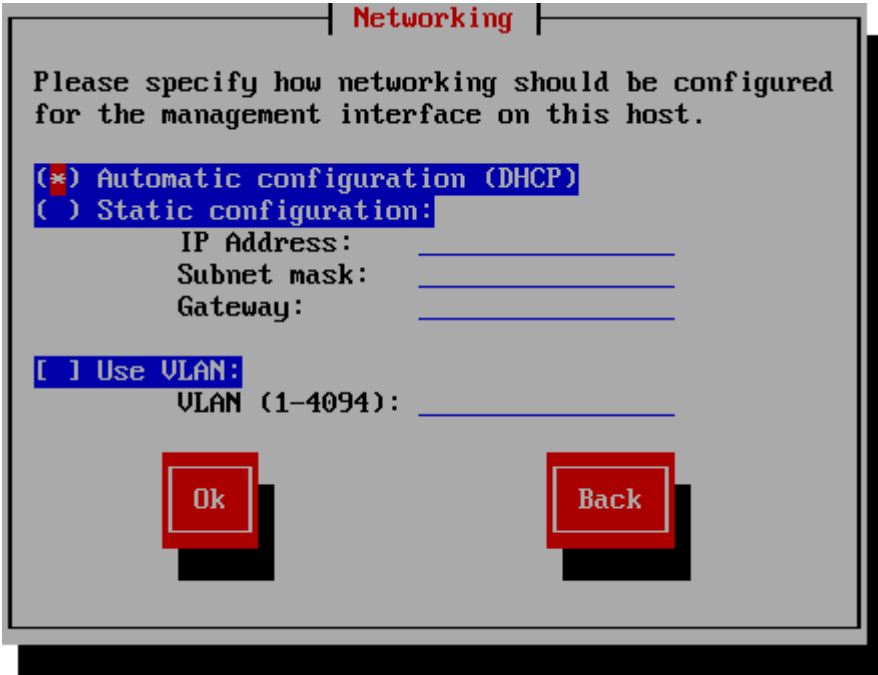


Figure 28:XCP Installation source

Select local media and it will find the ISO that is on the thumb stick, and this will be used to install XCP

After you can skip verification.

After we will select our network settings



The image shows a 'Networking' configuration window. It asks the user to specify how networking should be configured for the management interface. There are two main options: 'Automatic configuration (DHCP)' which is selected with a red asterisk, and 'Static configuration' which is unselected. Under 'Static configuration', there are input fields for 'IP Address', 'Subnet mask', and 'Gateway'. Below these is an option '[] Use VLAN:' with a corresponding 'VLAN (1-4094):' input field. At the bottom, there are two red buttons labeled 'Ok' and 'Back'.

Networking

Please specify how networking should be configured for the management interface on this host.

(*) Automatic configuration (DHCP)

() Static configuration:

IP Address: _____

Subnet mask: _____

Gateway: _____

[] Use VLAN:

VLAN (1-4094): _____

Ok Back

Figure 29:XCP network configuration

Then enter your desired password and confirm it for the root user on XCP-ng



The image shows a 'Set Password' configuration window. It asks the user to specify a password of at least 6 characters for the root account. A note states: '(This is the password used when connecting to the XCP-ng Host from XCP-ng Center.)'. There are two input fields, one for 'Password' and one for 'Confirm', both masked with asterisks. At the bottom, there are two red buttons labeled 'Ok' and 'Back'.

Set Password

Please specify a password of at least 6 characters for the root account.

(This is the password used when connecting to the XCP-ng Host from XCP-ng Center.)

Password: *****

Confirm: *****

Ok Back

Figure 30:XCP password

Then enter your desired DNS servers or leave at default which we will do for this example.

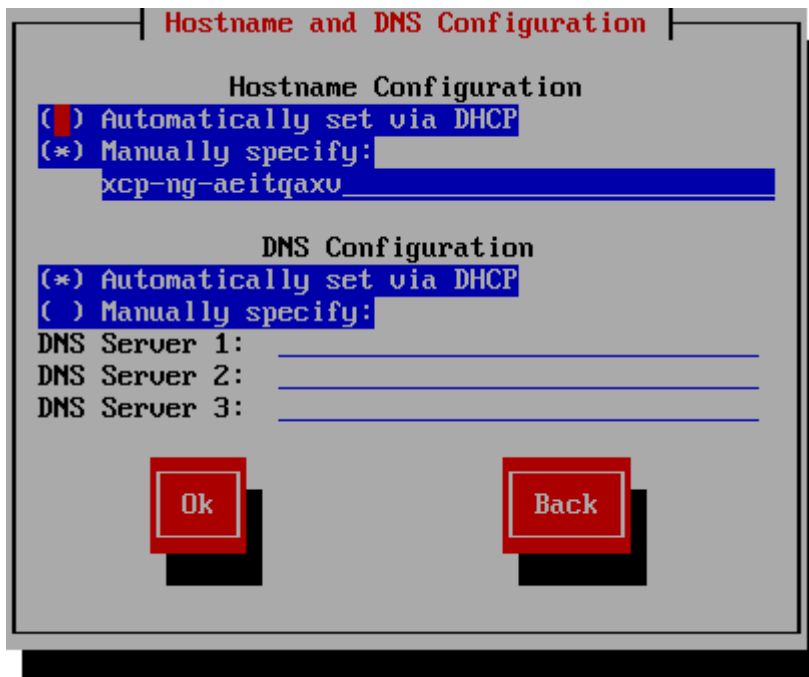


Figure 31:XCP hostname and dns

The next steps will be selection of region and time zone: select your desired for both. And lastly you will see a screen that says install xcp and select install: this will finalize the installation settings and will begin installing for the next 10-15 minutes.

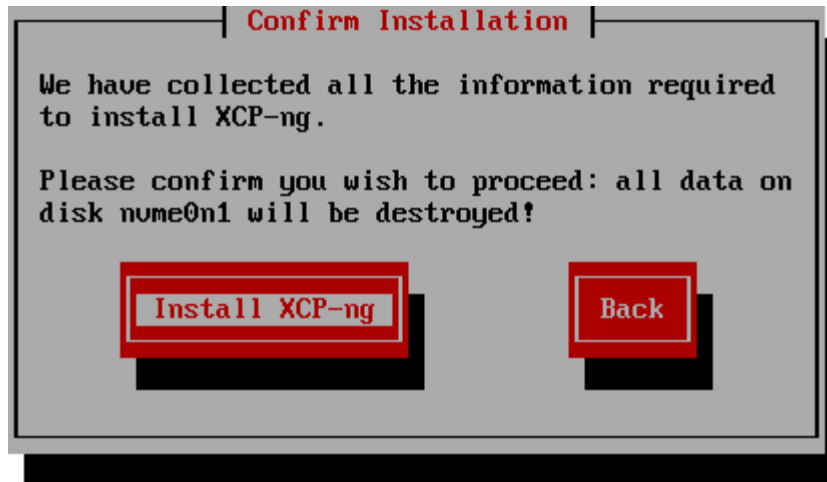


Figure 32: Confirm installation

After the installation finishes it will prompt you to install additional packages simply select “No”

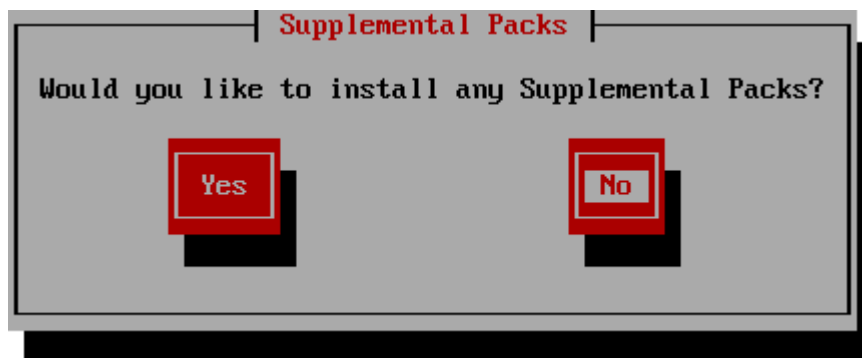


Figure 33: Supplemental packs prompt

After this finishes the screen will prompt you to reboot and launch itself into XCP-ng.

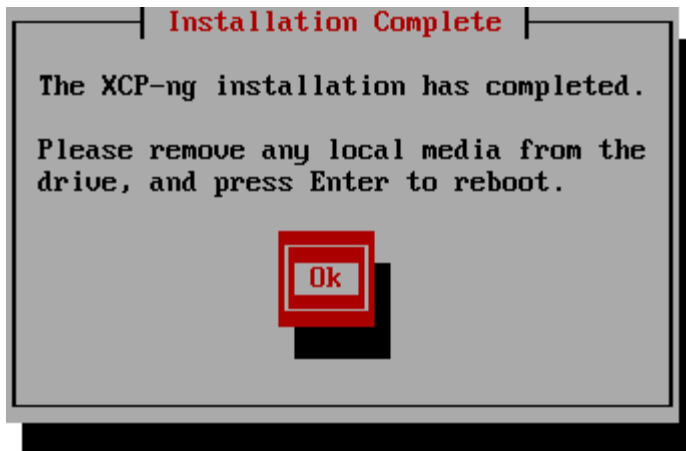


Figure 34:XCP Installation Complete

XCP-ng Initial Setup

Step 1. Xen Orchestra

Once XCP-ng has been installed and booted up on your machine you should see this following screen.

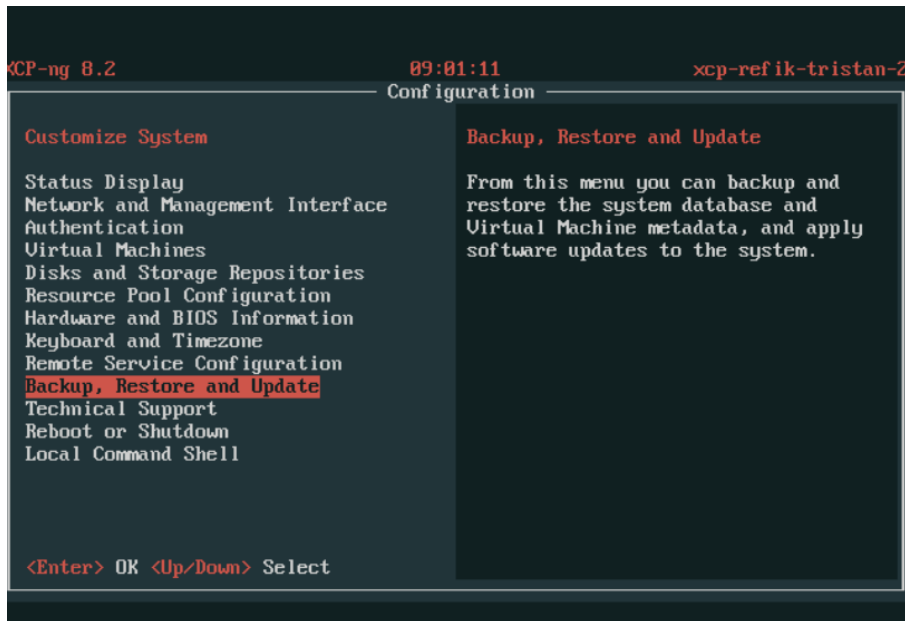
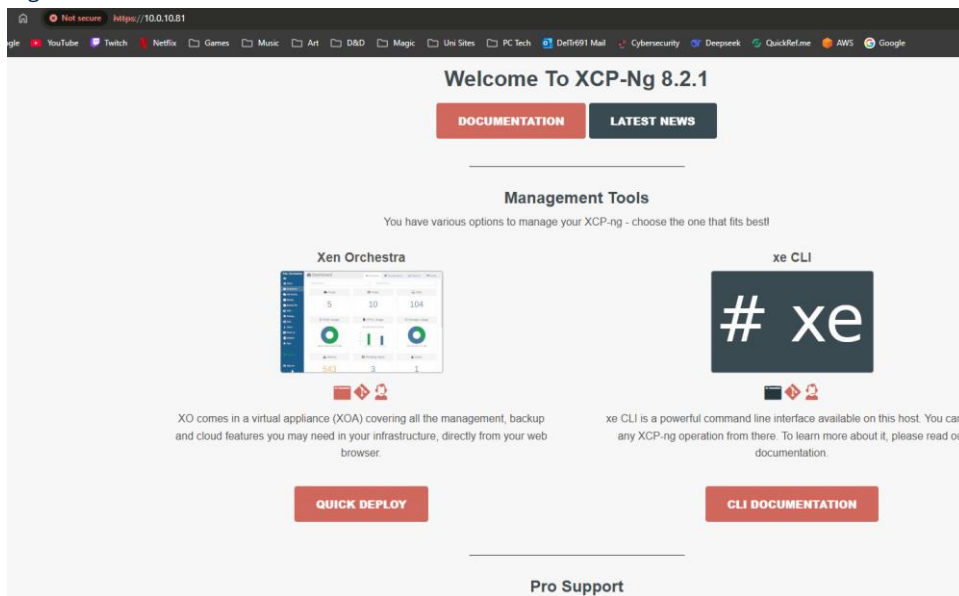


Figure 35:XCP main screen

From here you can see and operated almost all of XCP-ng however we will first be needing to setup the Xen Orchestra which is the virtual machine tool that governs XCP modules through its own web portal. Once this screen is visible XCP-ng should be live so connect to the server IP using the IP you entered in the previous step for the XCP-ng Server.

You will be prompted with the following screen:

Figure 36: Web UI Welcome Screen



From here you click on the quick deploy button and this will prompt you to enter the information requirements for the Xen Orchestra VM to manage the XCP through its web ui.

Figure 38: VM Managenet Config

Xen Orchestra Quick Deploy

This will download and deploy a new virtual machine on your host, called "XOA" (Xen Orchestra virtual Appliance).

Select a storage
Local storage - 167 GiB left

Select a network
Pool-wide network associati

XOA VM IP address

Netmask
255.255.255.0

Gateway

DNS
8.8.8.8

Leave everything empty to use DHCP. Provide all the information otherwise.

NEXT

[Close](#)

Firstly select the drive you want the VM to operate on

Then you select the network module it will be attached to

Then you enter the desired static ip of the machine for connection

Then you enter the network information such as subnet mask if applicable and gateway and DNS

After completion select deploy

This will create the XEN Orchestra VM hosted on your server at the input IP

Step 2. Creation of SR or Storage Remotes

For the creation of VMs on our server we will first need to add our storage to the XCP server so that it can pull the ISOs from our TrueNAS.

This is done with the following process. Connect to the web UI for Xen Orchestra:

Under the left hand panel in the New tab select new Storage:

Here you will see the following screen where you will input all the information about the share then click create:

Select the following parameters

- Select the server which will be attached to the storage,.
- Select an appropriate name and description.
- Select ISO for ISO file storage or ext local for just a local VDI.
- For the Share you need to input the domain of the server that is hosting the share.
- Then the credentials for the server.

Figure 39:XCP SR creation

Xen Orchestra

Create a new SR

General

Host: xcp-refik-tristan-2 (26.72 GiB RAM free) - xcp-refik-tristan-2

Name:

Share:

Description: TrueNAS Share

Select storage type: SMB ISO

Settings

Server: \\10.0.12.55\isos

User name: tristan

Password:

Storage usage

Summary

Name:

Share:

Description: TrueNAS Share

Type:

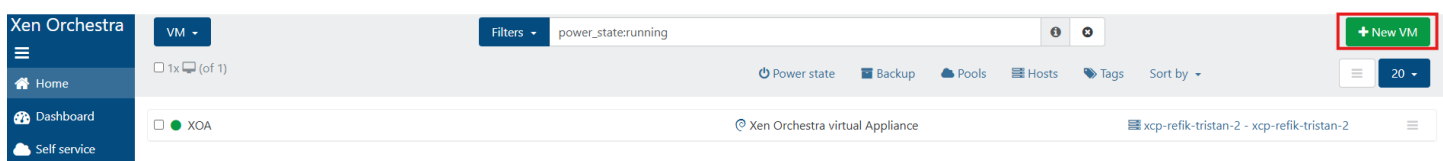
Once done click create and your storage will be created.

Step 3. VM Creation

For creating a VMs we first need the storage setup for where we will be pulling the ISOs from and we already did that in step 2.

Navigate to the panel that is labelled VM and within this menu click new VM on the top right.

Figure 40: Creation of VM



From this you will get to another screen and here we need to fill out some information before creating a VM.

- Firstly you select the Server the VM will be hosted on.
- Then you select a template (This is more flavour and does nothing other than select a preconfigured set of ram and cpus based on the template)
- Then you select the ISO which we will use the ISO from our TrueNAS share.
- Then we attach ethernet module and storage to the VM if applicable.

Figure 41: Creation of VM advance

Create a new VM on xcp-refik-tristan-2

- Info**

Template: Ubuntu Bionic Beaver 18.04 - ... Name: Ubuntu Bionic Beaver 18.0 Description: Ubuntu
- Performance**

vCPUs: 1 RAM: 1 GiB Topology: 1 socket with 1 core per socket
- Install settings**

☐ PXE
☒ ISO/DVD: guest-tools.iso - XCP-ng Tools
- Interfaces**

MAC: Auto-generated if empty Network: Pool-wide network associated ...
+ Add interface
- Disks**

SR: Local storage (146.54 GiB free ...) Name: Ubuntu Bionic Beaver 18.0 Description: Created by XO
Size: 10 GiB
+ Add disk
- Advanced**

Show advanced settings

- After we click on create VM at the bottom of the page and the vm will be created and started

After Creation your VMs will be listed in the VM section of the VMs tab. From here you can select your VMs and issue commands such as “start” and “stop”.

Figure 42: After creation of VM

VM			Filters	Power state	Backup	Pools	Hosts	Tags	Sort by	+ New VM
3x (of 3)										
☐	Ubuntu Jammy Jellyfish 22.04	Ubuntu Server	xcp-tristan-refik-2 - xcp-tristan-refik-2							
☐	Windows Server 2019 (64-bit)	Win Server 2K19	xcp-tristan-refik-2 - xcp-tristan-refik-2							
☐	XOA	Xen Orchestra virtual Appliance	xcp-tristan-refik-2 - xcp-tristan-refik-2							

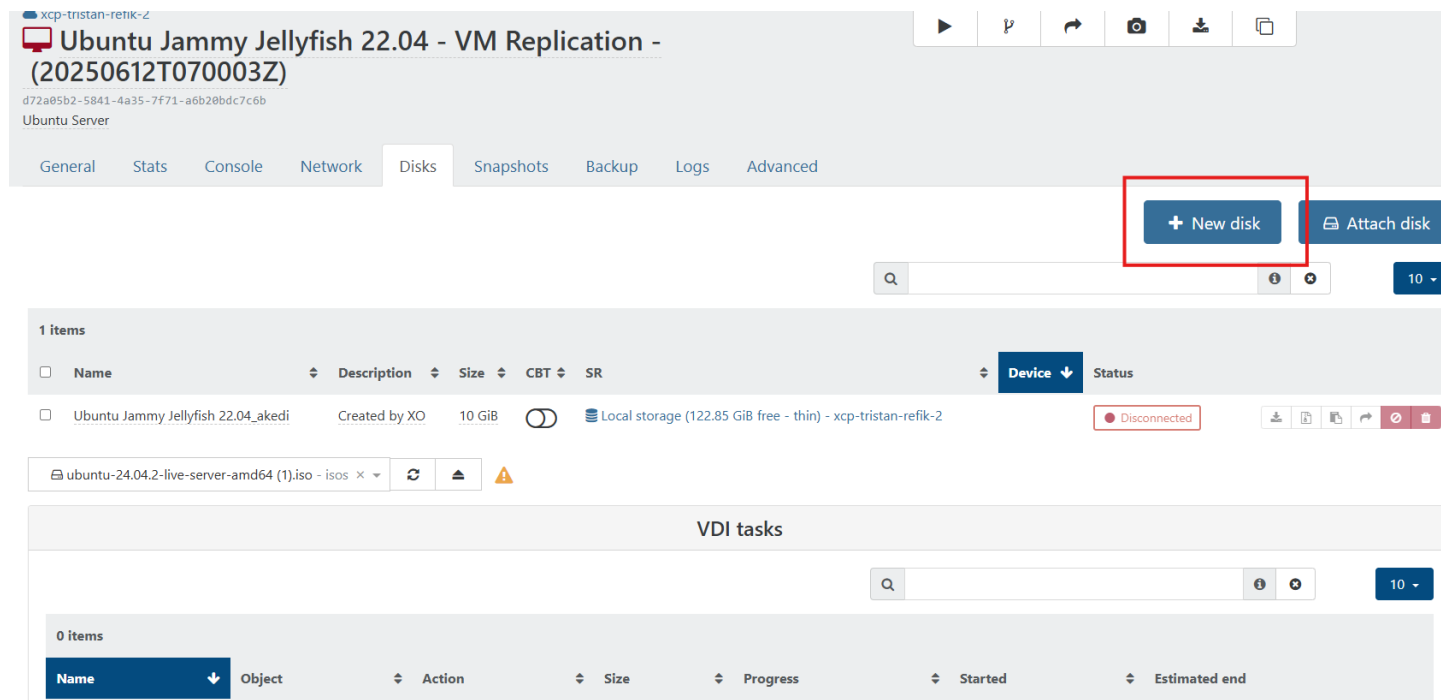
Step 4. VM elements configuration

After or during the VM creation process you have the ability to allocate more storage, ram, CPUs to the VM. In the VM screen you can select a VM and enter the VM overview screen in which here you see multiple tabs.

Step 4.1 Storage Allocation to VMs

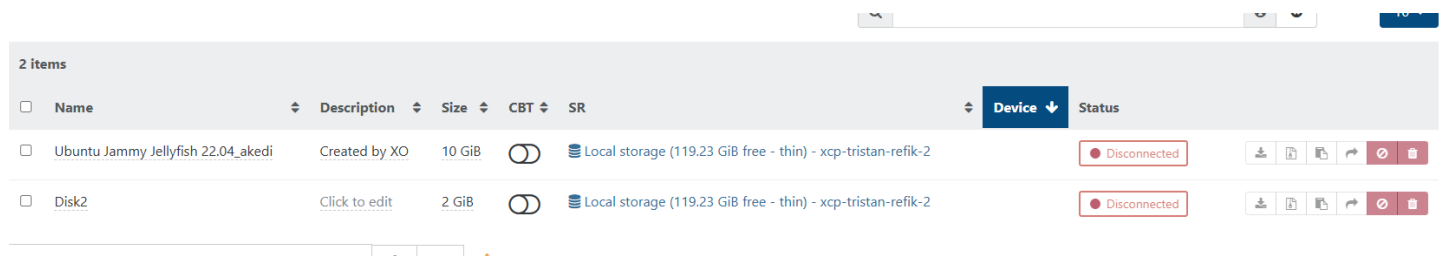
In the VM general screen click on the Disk tab.

Figure 43:ADD Disk xcp



Here you will click on the New disk button with which you can specify a new VDI to create and then after you can select it in the attach Disk option to mount It to the current VM.

Figure 44:After disk creation

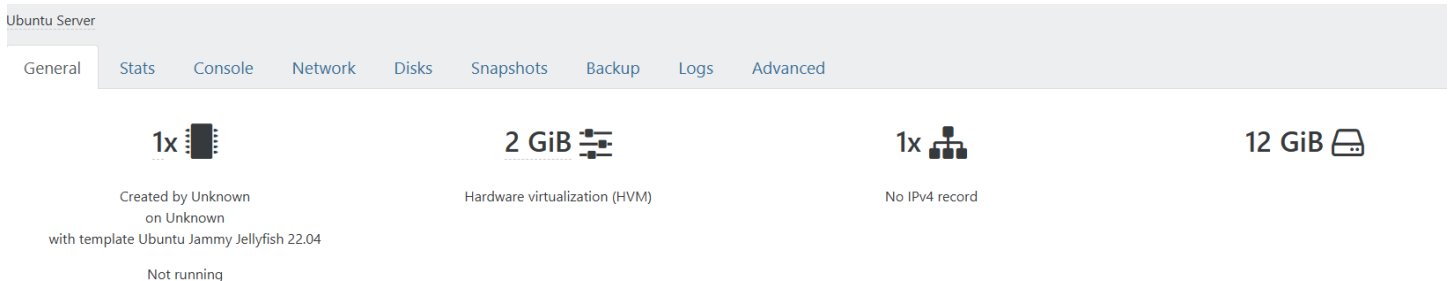


As you can see now I have added a new disk and it will be mounted once you click on the disconnected button. If you rather detach a drive you can click on the same status button the switch between mounted and unmounted functions.

Step 4.2 Ram & CPUs Allocation

These are relatively simpler than storage here you can just navigate to the Main general tab and double click either the Ram or the CPUs number and specify another number that will change the allocated ram in Gb or CPUs. This action may only be done when the machine is turned off.

Figure 45: RAM AND CPU Allocation



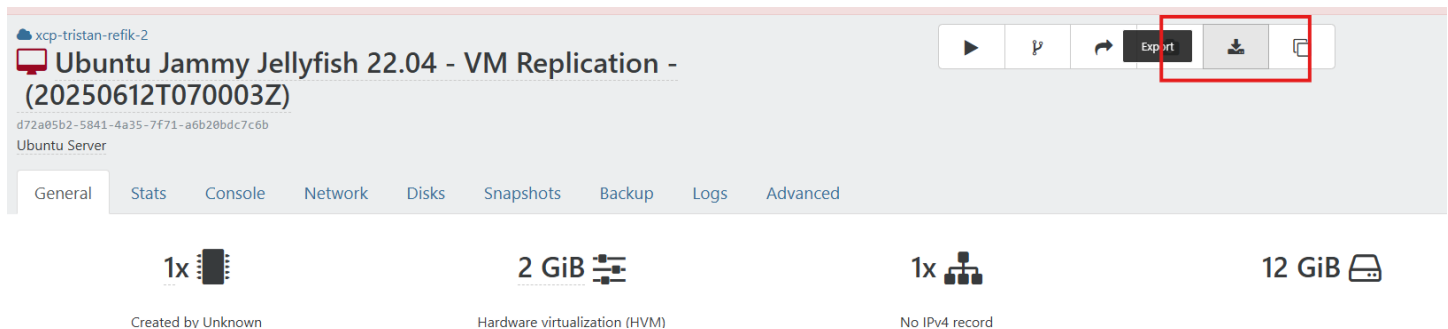
5. VM Export

In this section I will outline how to export a VM for use somewhere else if needed.

Navigate to the general VMs tab and select your desired VM.

In the top right hand panel, you click on a button called Export. This will open the export wizard and will download the VM into a zip file on your current machine.

Figure 46: VM Export



6. Templates

Within XCP-ng you can create templates of Vms for quick and easy creation. Firstly navigate to the dashboard and select the templated tab. From here you will click the new button on the top right hand corner.

Figure 47: Templates

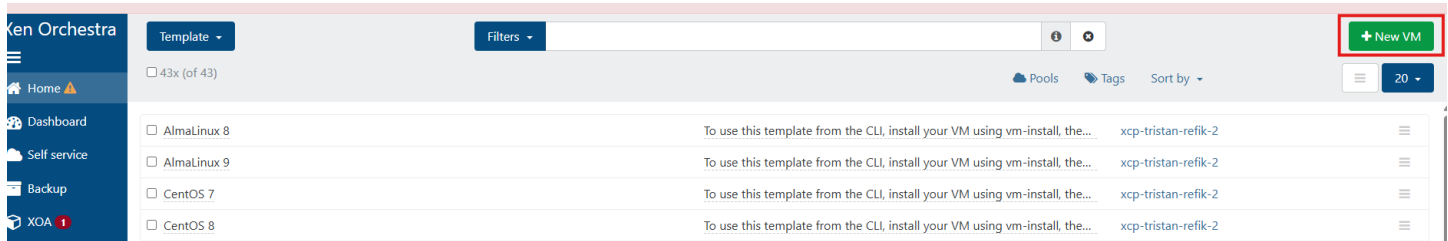


Figure 48: Creation of VM with templates

Create a new VM on xcp-tristan-refik-2

- Info**
 - Template: Ubuntu Jammy Jellyfish 22.04
 - Name: Ubuntu 20.04 Template
 - Description: Ubuntu Template
- Performance**
 - vCPUs: 1
 - RAM: 2.5 GiB
 - Topology: 1 socket with 1 core per socket
- Install settings**
 - ☐ PXE
 - ☒ ISO/DVD: ubuntu-24.04.2-live-server-am...
- Interfaces**
 - MAC: Auto-generated if empty
 - Network: Pool-wide network associated ...
 - [+ Add interface](#)
- Disks**
 - SR: Local storage (119.23 GiB free ...)
 - Name: Ubuntu Jammy Jellyfish 22
 - Description: Created by XO
 - Size: 10 GiB
 - [+ Add disk](#)
- Advanced**
 - [Show advanced settings](#)
- Summary**
 - 1x
 - 2.5 GiB
 - 1x
 - 1x

[Reset](#) [Create](#)

This will open up the template wizard which will allow you to define the parameters for your desired template and create it.

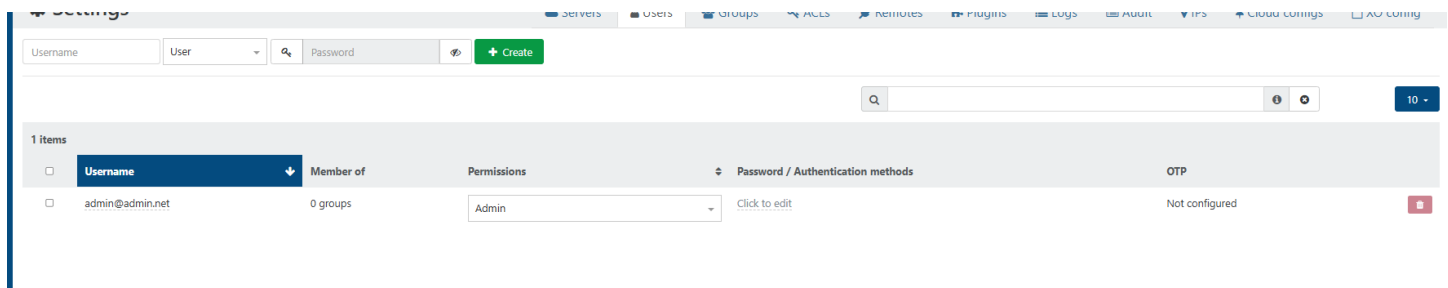
From the template now you can select when creating new VM in the templates tab.

7. Managing Users and Permissions

To manage users in XCP you need to navigate to the users tab under settings.

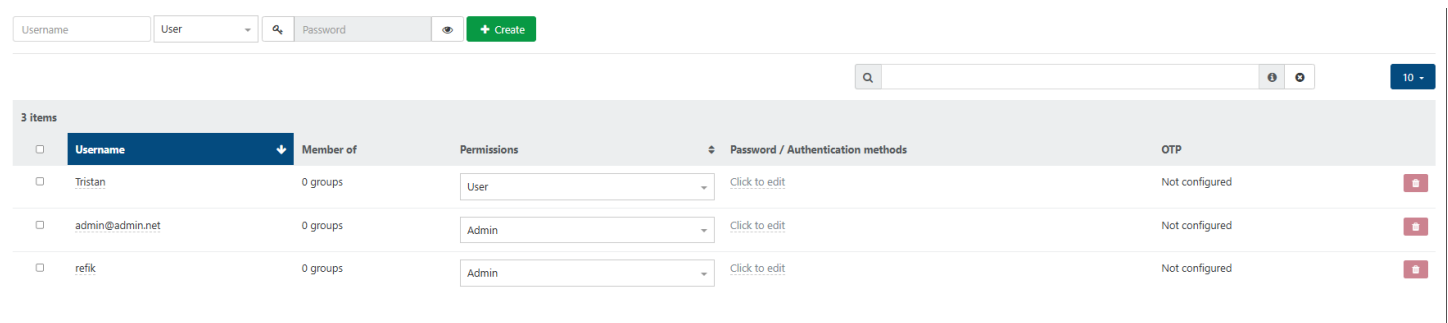
Step 7.1 Users

Figure 49: User managing



Here you can click on the create to create a user within XCP server.

Figure 50: Creation of users



After you input your desired username and password and their respective role you can create a user and it will be added to the main users list below.

In the overview you will see all users and their respective groups, permissions role, password and ability to adjust if lost, and their current OTP in case of use of an authenticator.

Step 7.2 Groups

After the users have been created, we can manage the groups that they are associated with to make management more fluid and dynamic. Within the Groups tab we can also in the top left hand corner add a new group name and click new to add it to the list.

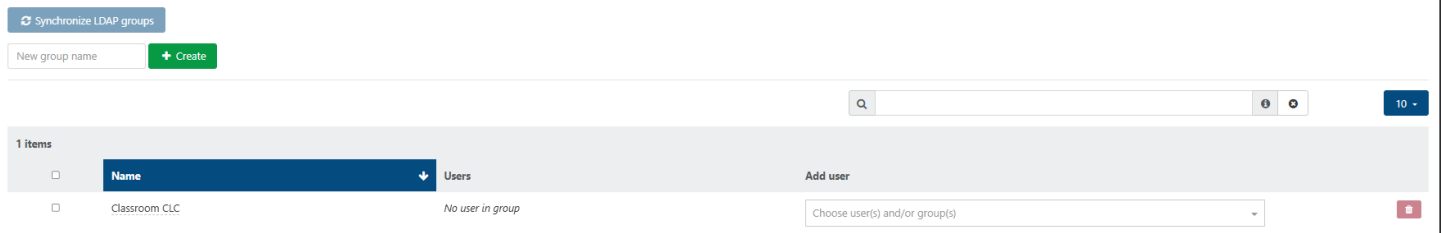


Figure 51: Creation of Classroom CLC Group

Step 7.3 Access Control Lists

Here in the next tab we have the ACLs, this is where we can define some permissions for users and groups in accordance with our Virtual Machines and Backup Jobs.

So as you can see in the following screenshot we have added the Classroom CLC to have the role of admin on the ubuntu VM and we can add tags like the continuous replication tab

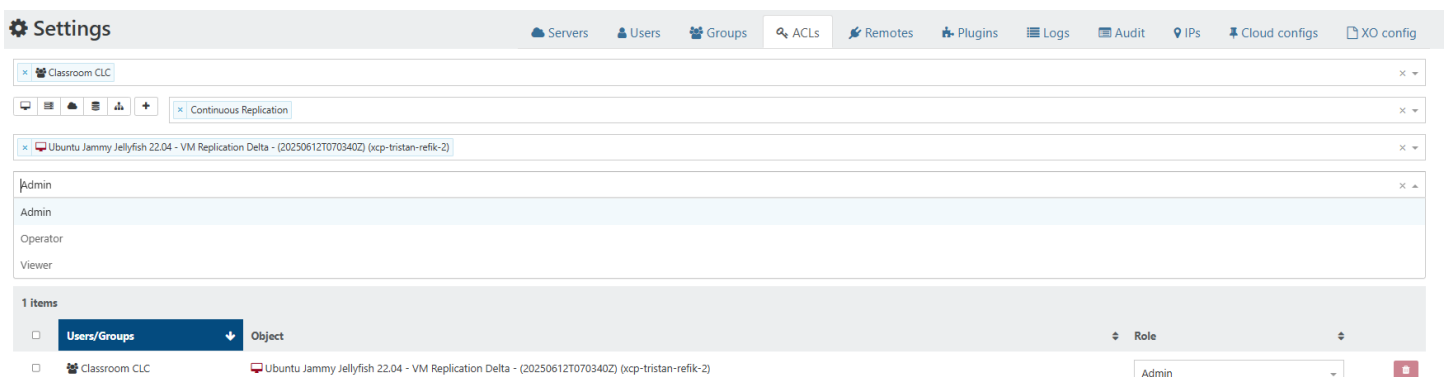


Figure 52: ACL for classroom group

This will add all users of the classroom CLC group the admin role on our ubuntu VMs.

8. XCP-ng Updates & Patching

To update XCP you first click on the XOA tab and select the update section. Here you will see this screen.

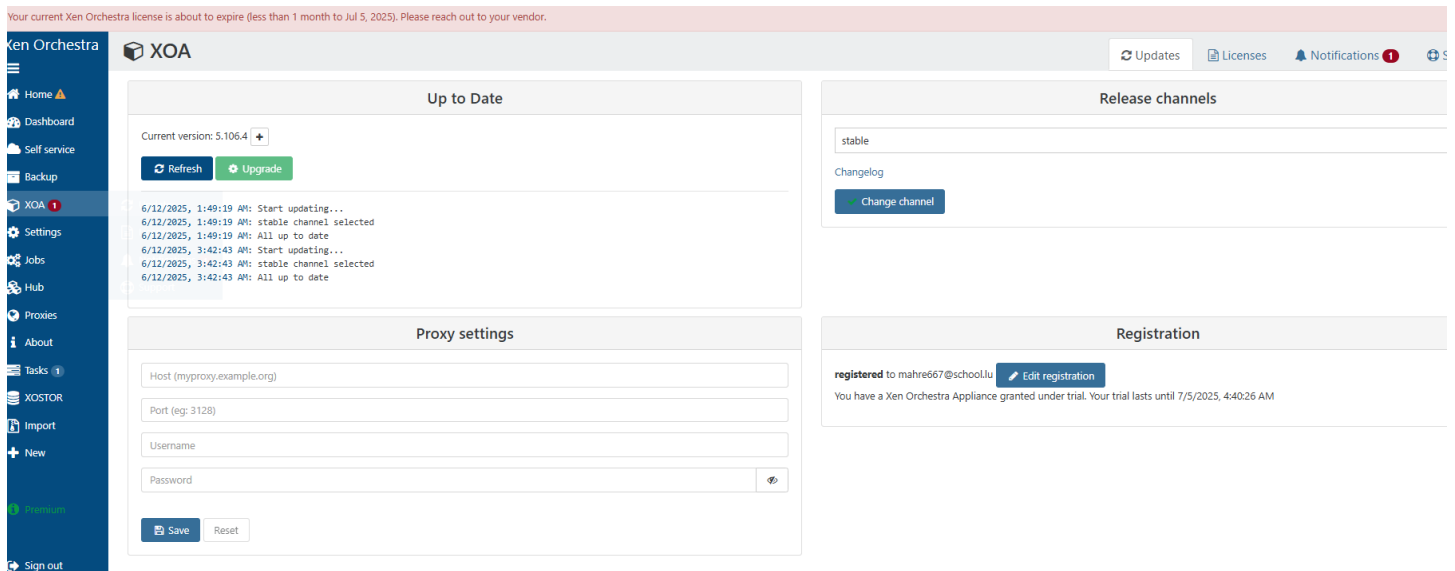
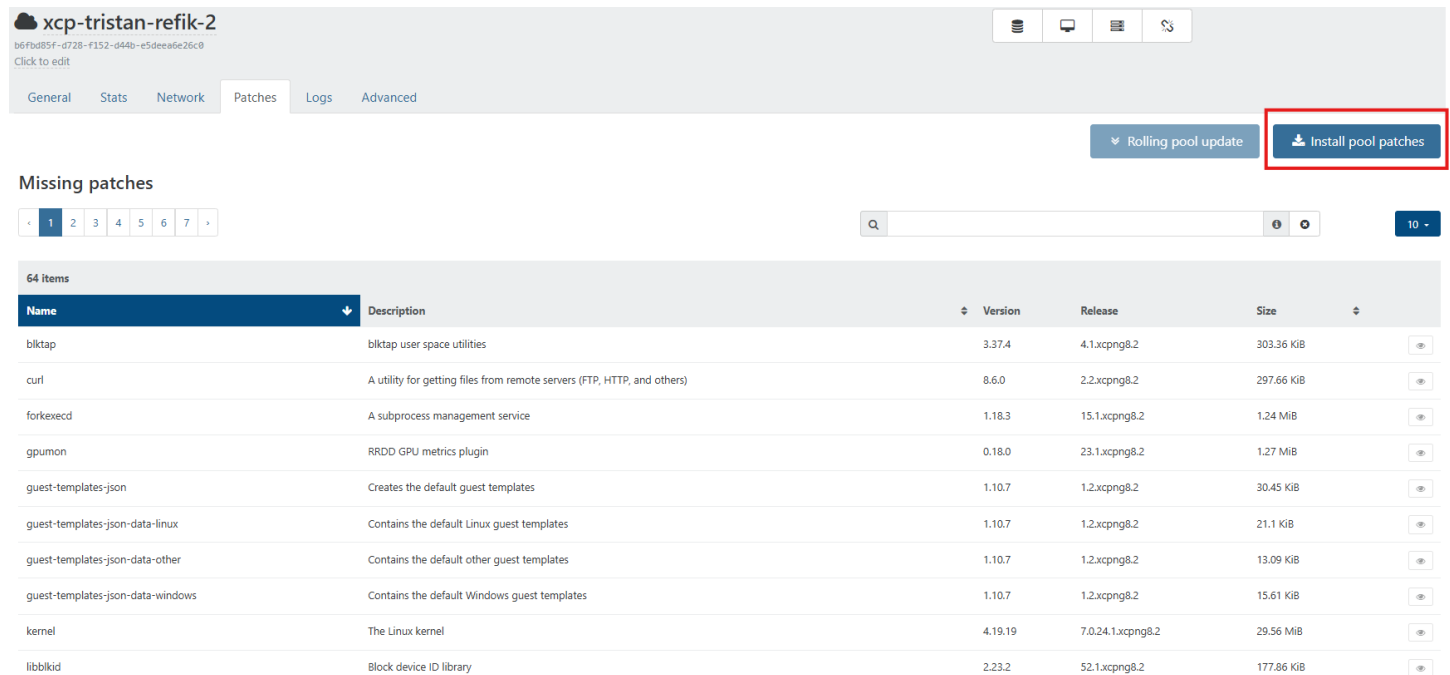


Figure 53: Update screen XCP

Here you will need to login to your Xen Orchestra account to validate your subscription as certain updates and patches are only available to certain subscription types. After your license and subscription have been validated you can click the Update button on the top left and after the update has finished you can click upgrade. This will update all your XOA and XCP-ng Modules and files to the latest versions.

8.1 Patching

To get to the patching screen of the server you will first navigate to the Home tab, and the Pools subsection tab, then the Patches subsection tab within the xcp-ng server. From here you can see all missing patches and all current patches listed.



The screenshot shows the xcp-ng management interface for a server named 'xcp-tristan-refik-2'. The 'Patches' tab is selected, displaying a list of missing patches. In the top right corner, the 'Install pool patches' button is highlighted with a red rectangular box. Below the button, the 'Missing patches' section shows a table of 64 items.

Name	Description	Version	Release	Size
blktap	blktap user space utilities	3.37.4	4.1.xcpng8.2	303.36 KiB
curl	A utility for getting files from remote servers (FTP, HTTP, and others)	8.6.0	2.2.xcpng8.2	297.66 KiB
forkexecd	A subprocess management service	1.18.3	15.1.xcpng8.2	1.24 MiB
gpumon	RRDD GPU metrics plugin	0.18.0	23.1.xcpng8.2	1.27 MiB
guest-templates-json	Creates the default guest templates	1.10.7	1.2.xcpng8.2	30.45 KiB
guest-templates-json-data-linux	Contains the default Linux guest templates	1.10.7	1.2.xcpng8.2	21.1 KiB
guest-templates-json-data-other	Contains the default other guest templates	1.10.7	1.2.xcpng8.2	13.09 KiB
guest-templates-json-data-windows	Contains the default Windows guest templates	1.10.7	1.2.xcpng8.2	15.61 KiB
kernel	The Linux kernel	4.19.19	7.0.24.1.xcpng8.2	29.56 MiB
libblkid	Block device ID library	2.23.2	52.1.xcpng8.2	177.86 KiB

Figure 54: Manual Install Patches Wizard

Here in the top right hand side you can click on the Install pool patch and there your can input the parameter in the wizard for the patch you would like to install.

12728fa8-4dda-4abb-9478-2579512f37af
Default install

General Stats Console Network Storage Patches **64** Logs Advanced

[Install all patches](#)

< 1 2 3 4 5 6 7 >

Q 10 -

64 items

Name	Description	Version	Release	Size
blktap	blktap user space utilities	3.37.4	4.1.xcpng8.2	303.36 KiB
curl	A utility for getting files from remote servers (FTP, HTTP, and others)	8.6.0	2.2.xcpng8.2	297.66 KiB
forkexecd	A subprocess management service	1.18.3	15.1.xcpng8.2	1.24 MiB
gpumon	RRDD GPU metrics plugin	0.18.0	23.1.xcpng8.2	1.27 MiB
guest-templates-json	Creates the default guest templates	1.10.7	1.2.xcpng8.2	30.45 KiB
guest-templates-json-data-linux	Contains the default Linux guest templates	1.10.7	1.2.xcpng8.2	21.1 KiB
guest-templates-json-data-other	Contains the default other guest templates	1.10.7	1.2.xcpng8.2	13.09 KiB
guest-templates-json-data-windows	Contains the default Windows guest templates	1.10.7	1.2.xcpng8.2	15.61 KiB
kernel	The Linux kernel	4.19.19	7.0.24.1.xcpng8.2	29.56 MiB
libblkid	Block device ID library	2.23.2	52.1.xcpng8.2	177.86 KiB

Figure 55: Patch Screen in XCP Server

Furthurmore under the Hosts tab in the XCP-ng server subsection there is a missing patches prompt for your server and in the top right you can click install all patches this will notify you with a red indicator that you have X missing patches that can be installed.

Name	Description	Version	Release	Size
blktp	blktp user space utilities	3.37.4	4.1.xcpng8.2	303.36 KiB
curl	A utility for getting files from remote servers (FTP, HTTP, and others)	8.6.0	2.2.xcpng8.2	297.66 KiB
forkexecd	A subprocess management service	1.18.3	15.1.xcpng8.2	1.24 MiB
gpumon	RRDD GPU metrics plugin	0.18.0	23.1.xcpng8.2	1.27 MiB
guest-templates-json	Creates the default guest templates	1.10.7	1.2.xcpng8.2	30.45 KiB
guest-templates-json-data-linux	Contains the default Linux guest templates	1.10.7	1.2.xcpng8.2	21.1 KiB
guest-templates-json-data-other	Contains the default other guest templates	1.10.7	1.2.xcpng8.2	13.09 KiB
guest-templates-json-data-windows	Contains the default Windows guest templates	1.10.7	1.2.xcpng8.2	15.61 KiB
kernel	The Linux kernel	4.19.19	7.0.24.1.xcpng8.2	29.56 MiB
libblkid	Block device ID library	2.23.2	52.1.xcpng8.2	177.86 KiB

Figure 56: Missing Patches indicator

Specialization: Backups

This will be the section within our documentation that outlines the entire roster of tools and utilities that come with XCP-ng that allow for data retention e.i “Backups”, “Replications”, “Restoration”.

Since this is the specialization, we will go into very deep detail about every possible feature present within XCP-ng. With this in mind some features will be unavailable to us due to them being limited to only paid membership.

Part 1. Backups

Within XCP-ng with the first tier of their subscription you can create a backup of VMs within your server. just navigate to the Backups tab under the Home Section.

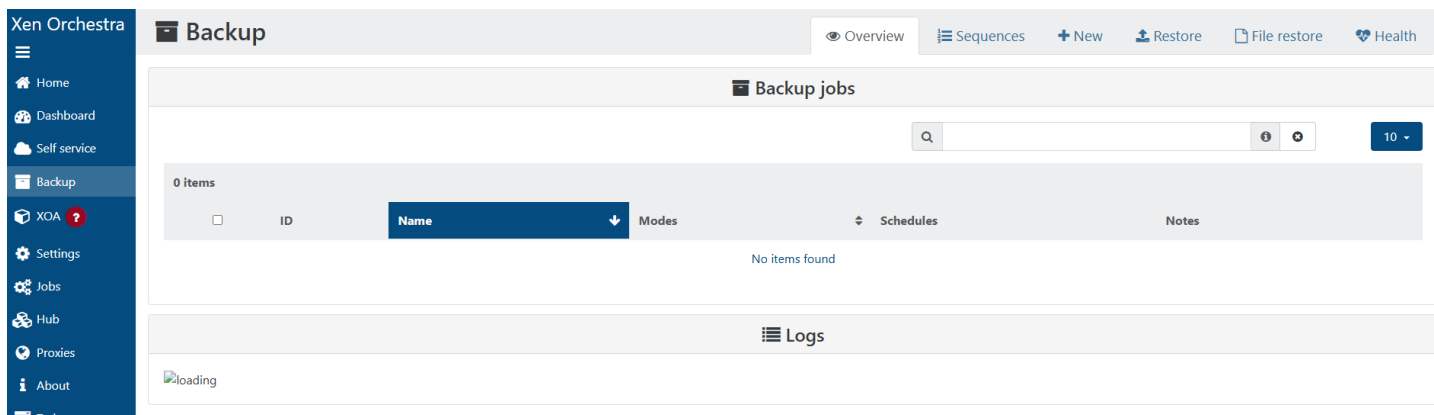


Figure 57: Backups Overview Tab

Within this Overview tab you will see your current backup jobs which we have none and any backup logs which we also have none.

Up on the right-hand side we can see there are 4 tabs open for sequences, New, Restore, File Restore and Health.

Part 2. Health

Starting with the simplest tab first we will cover the Health tab. The health tab is more of a individual monitoring of backup or snapshots done that do not fall under the criteria of being scheduled or planned in any way. Within this tab we can see:

All data within these tabs is structures with a Creation date, Name of backup or Snapshot, Description noted, and Snapshot or backup device (Either name of VM or location of storage)

- **Detached backups:** These are backups that have been processed independently from a backup job or sequence meaning they most likely have been done manually by an individual. And we can refresh the tab to see if there were any backups made more recently.
- **Detached VM Snapshots:** Same as before these are snapshots of a Virtual Machine that has been done outside of a regular job or sequence, and this is for better monitoring of important snapshots.
- **Legacy Backups Snapshots:** This tab outlines backups and snapshots that have been outdated format or have been superseded by another existing backup that

retains all its data and thus rendering the legacy backup pointless. This is to monitor your backups health and maintain and more organized layout for backups and help reduce clutter or overreplication.

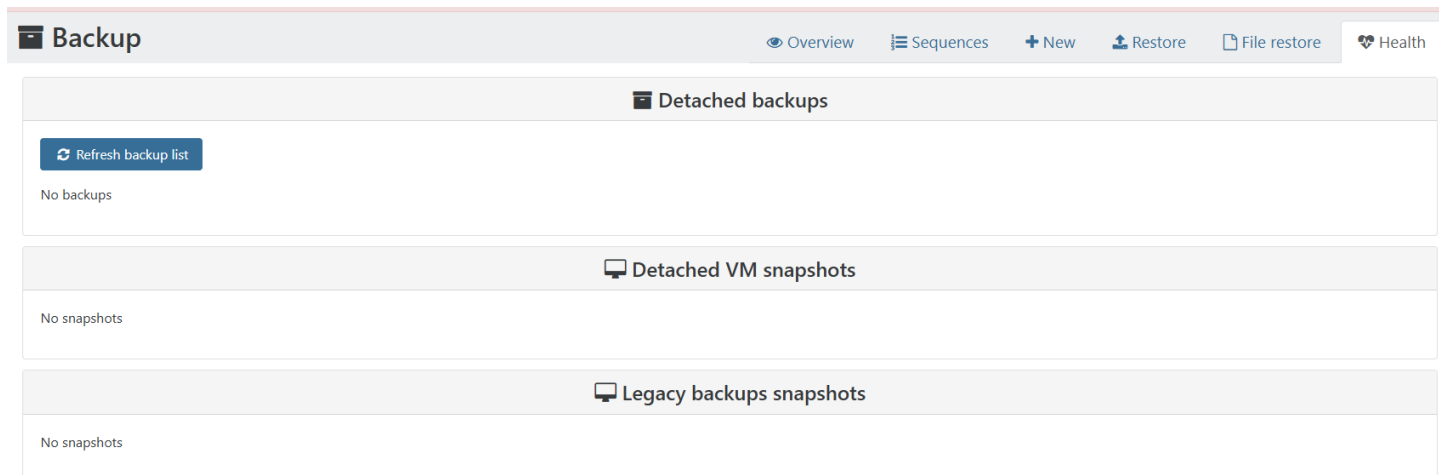


Figure 58: Health Main Tab

Part 3. New Tab

Within the New tab we have all the listed options for new backup jobs. These are the options listed.

VM Backup & Replication: Standard option for simple backup types and replication type jobs.

VM Mirror Backups: This will create a 1 to 1 identical copy of the VM that will be used for high availability incase operations fail on one VM.

XO Configs & Pool Meta Data Backup: This will just have two operations to backup the Xen Orchestra Configs and to back up the Servers Metadata

Sequences: This is a set of defined steps that you can use in conjunction to create more complex backup solutions.

The screenshot displays the 'Backups Main Tab' interface. It features several sections for configuring backup jobs:

- Name***: A text input field for naming the backup job.
- VMs to backup***: A dropdown menu to select the VM(s) to be backed up, with a 'Smart mode' toggle.
- Schedules***: A section for scheduling backups, currently showing 'No schedules found' and a '+' button to add new schedules.
- Settings**: A section for advanced settings, including:
 - Proxy**: A dropdown menu to select a proxy.
 - Report when**: A dropdown menu with options like 'Skipped and failure', and a link to 'Plugins settings'.
 - Report recipients**: A text input field for email addresses, with a '+' button to add more recipients.
- Backup Methods**: A row of buttons for different backup methods: 'Rolling Snapshot', 'Backup', 'Delta Backup', 'Disaster Recovery', and 'Continuous Replication'. A note below states: 'Click for more information about the backup methods.'

Figure 59: Backups Main Tab

Here are the options listed. Note that some options may be selected together, and not all can be selected together as they would contradict each other.

Rolling snapshot: This will take a snapshot of the VM selected into the Storage Remote selected and with this it will rewrite the previous snapshot taken this is why its called a rolling snapshot.

Backup option: just a standard backup of the VM within the storage location selected.

Disaster Recovery: This name is rather odd as it is simply a replication job, this will replicate the VM to another location.

Delta Backup & Continuous Replicaiton jobs: These are both Incremental changes backups only they will either backup the VM or replicate the VM but only the changes done from the previous job.

Other Options within the tab include:

Proxy: You can specify a proxy that handles the backup stream as to not overload XOA in extreme cases or for higher stream bandwidth in case of bottlenecks.

Report when: Which specified when you want to be reported incase the job fails or is skipped or always or never.

Report recipients: this specifies the email addresses that the reports will be sent to.

Concurrency: this specifies how many of the jobs may be running at the same time.

Number of reties if VM fails: self-explanatory this will stop retying after the number input.

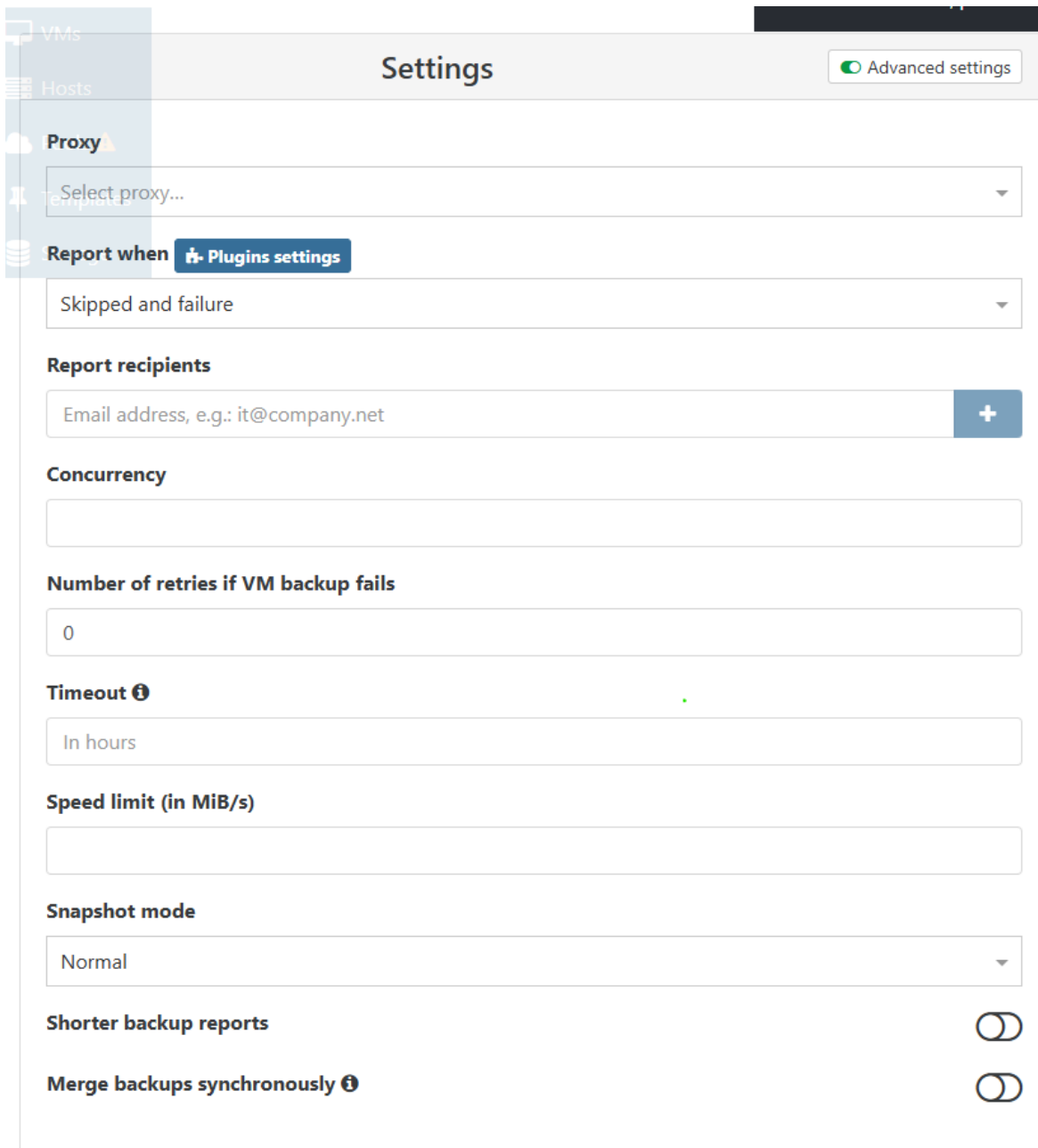
Timeout: time taken before job fails in case it takes too long or is stuck.

Speed limit: this limits the bandwidth in case bandwidth flooding is a problem.

Snapshot mode: This specifies either a full or incremental Snapshot that can be taken of the machine at the time of this job running.

Then you have the option to have **Shorter Reports** rather than full backup reports

And you have the option to **Enable merging backups synchronously**; this will make the merging of backups from incremental operate on multiple processes to prevent priority inversion or thread locking errors when chaining multiple backup processes.



Settings Advanced settings

Proxy

Select proxy...

Report when Plugins settings

Skipped and failure

Report recipients

Email address, e.g.: it@company.net +

Concurrency

Number of retries if VM backup fails

0

Timeout ⓘ

In hours

Speed limit (in MiB/s)

Snapshot mode

Normal

Shorter backup reports ⌵

Merge backups synchronously ⓘ ⌵

Figure 60: Backups Advanced Settings

Additionally with all Job type options you can specify a schedule in which the jobs will recur this will simply make a cronjob for the configuration.

🕒 Schedule sequence

ⓘ If your country participates in DST, it is advised that you avoid scheduling jobs at the time of change. e.g. 2AM to 3AM for US.

Month

January	February	March
April	May	June
July	August	September
October	November	December

1Every month

Hour

0	1	2	3	4	5
6	7	8	9	10	11
12	13	14	15	16	17
18	19	20	21	22	23

Every hour

Day

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

1Every day

Minute

0	1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18	19
20	21	22	23	24	25	26	27	28	29
30	31	32	33	34	35	36	37	38	39
40	41	42	43	44	45	46	47	48	49
50	51	52	53	54	55	56	57	58	59

Figure 61: backups Schedules

4. Restore Tab

Within the restore tab you will see the following items present:

Backup

Overview
Sequences
+ New
Restore
File restore
Health

Refresh backup list
Metadata

Explore the search syntax in the documentation
10

2 Items

☐	VM Name	VM Description	Oldest backup	Latest backup	Size	Available Backups	
☐	Ubuntu Jammy Jellyfish 22.04	Ubuntu Server	2025-06-11 15:01	2025-06-12 00:00	10.45 GiB	Full 2 Delta 1	☒ <input type="download"/> <input type="delete"/>
☐	Windows Server 2019 (64-bit)	Win server 2K19	2025-06-05 15:00	2025-06-05 15:00	7.59 GiB	Full 1	☒ <input type="download"/> <input type="delete"/>

Logs

10

1 Items

Job ID	Job name	Backup time	VM	Restore time	Duration	SR	Status	Size	Speed
0123	2025-06-11	Ubuntu Jammy Jellyfish 22.04 (20250611T220126Z) bco-	2025-06-11	2 minutes	Local storage (122.85 GiB free - thin)	Successful	3.42	30.31	GB/s

Figure 62: Backups Main tab After creation

Here you will see the VMs that you have within the primary tab and under there will be another table with the restores that have been done.

You can see that we have both our Windows and Ubuntu VMs backups with either separate Full or Delta Backups and we can select to restore from them within that list,

Once a backup has been completed it will be added to the restore logs at the bottom which we will have detailed information about restoration job that was processed.

After we click the restore button and select a full backup we can also always have the option to make a differential backup from a full if only certain files were affected.

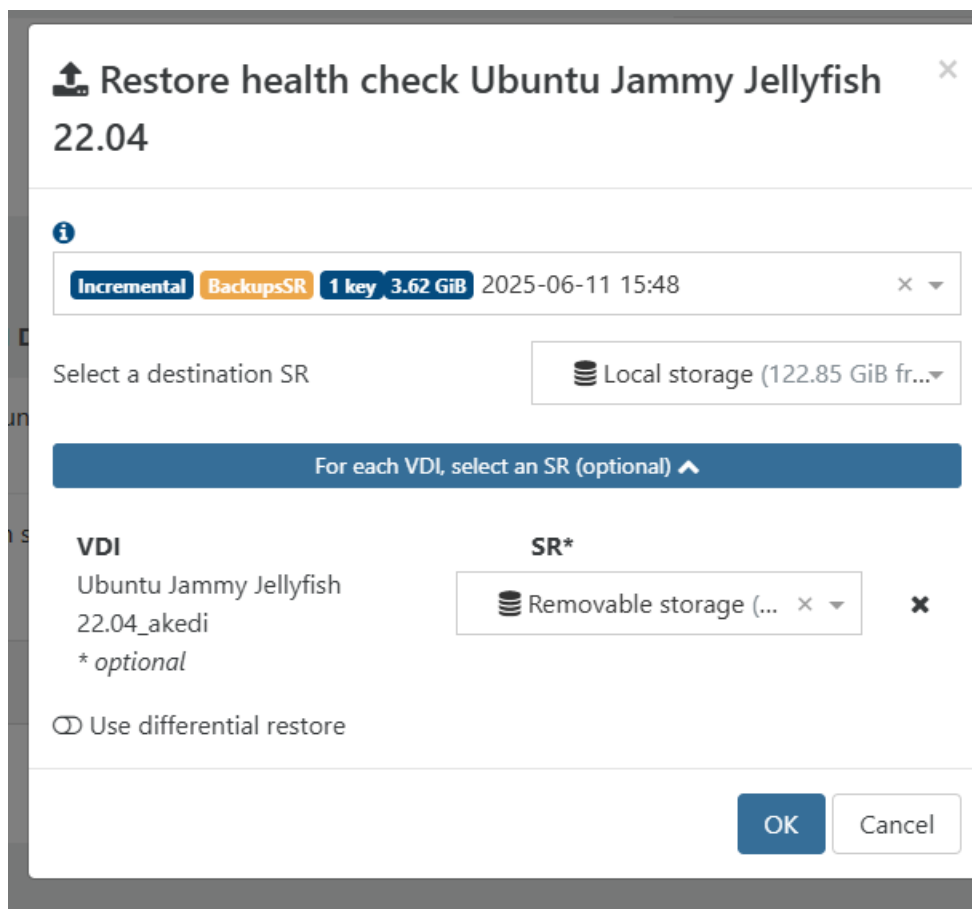


Figure 63: Backup restore wizard

Additionally, under the Logs tab you will see the listed information about the logs and you can also choose to open the logs raw file to inspect it manually.

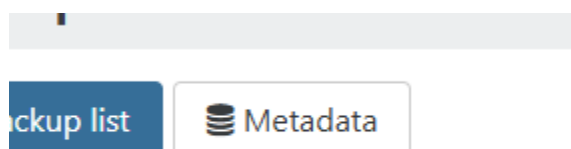
```

Pretty-print ☐
{
  "data": {
    "backupId": "0532ea68-bb2f-4d74-8be4-cc21e6d53993//xo-vm-backups/278d8c26-9ae6-792b-4b52-93132b4b96c9/20250611T220126Z.json",
    "jobId": "23190123-e67b-4c4d-b5ba-f409af05d67d",
    "srId": "8905780f-726e-26ee-973a-45c87373735b",
    "time": 1749679286683
  },
  "id": "1749681545274",
  "message": "restore",
  "start": 1749681545274,
  "status": "success",
  "tasks": [
    {
      "id": "1749681545284",
      "message": "transfer",
      "start": 1749681545284,
      "status": "success",
      "end": 1749681660657,
      "result": {
        "size": 3667022336,
        "id": "d23a46e4-6f05-b7b7-1052-2a67e87f3fff"
      }
    }
  ],
  "end": 1749681660658,
  "result": {
    "size": 3667022336,
    "id": "d23a46e4-6f05-b7b7-1052-2a67e87f3fff"
  }
}

```

Figure 64: Restore Raw File

Under the restore tab at the top left hand corner there is also another section for only metadata of the VMs, this button clicking on it will show you the logs of the backup jobs regarding the metadata.



5. File Restore

Here in this tab, you have the ability to manage the restoration of singular or many separate files. This tab will show you the VM name, The description of the VM, The Oldest backup, the latest backup and the available backups count.

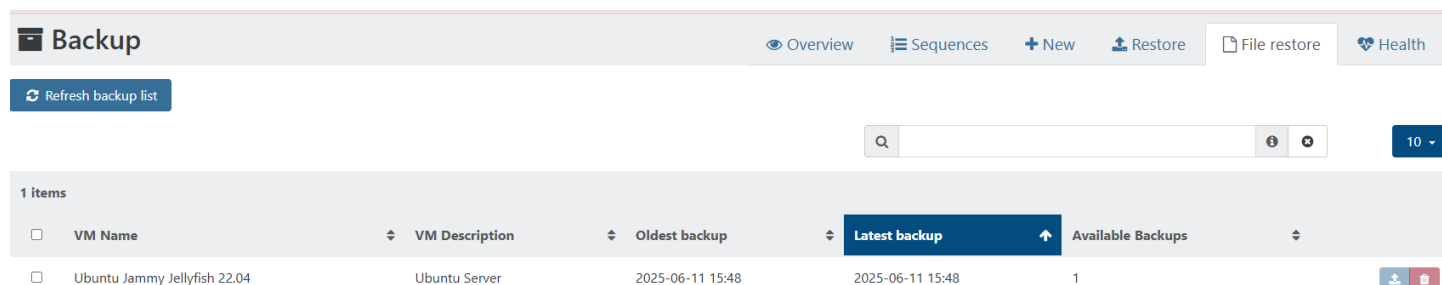


Figure 65: Main file restore Tab

From here you can select a Backup VM and click the restore, this will prompt you to open the backup ZIP and select the files you want to restore to the VM.

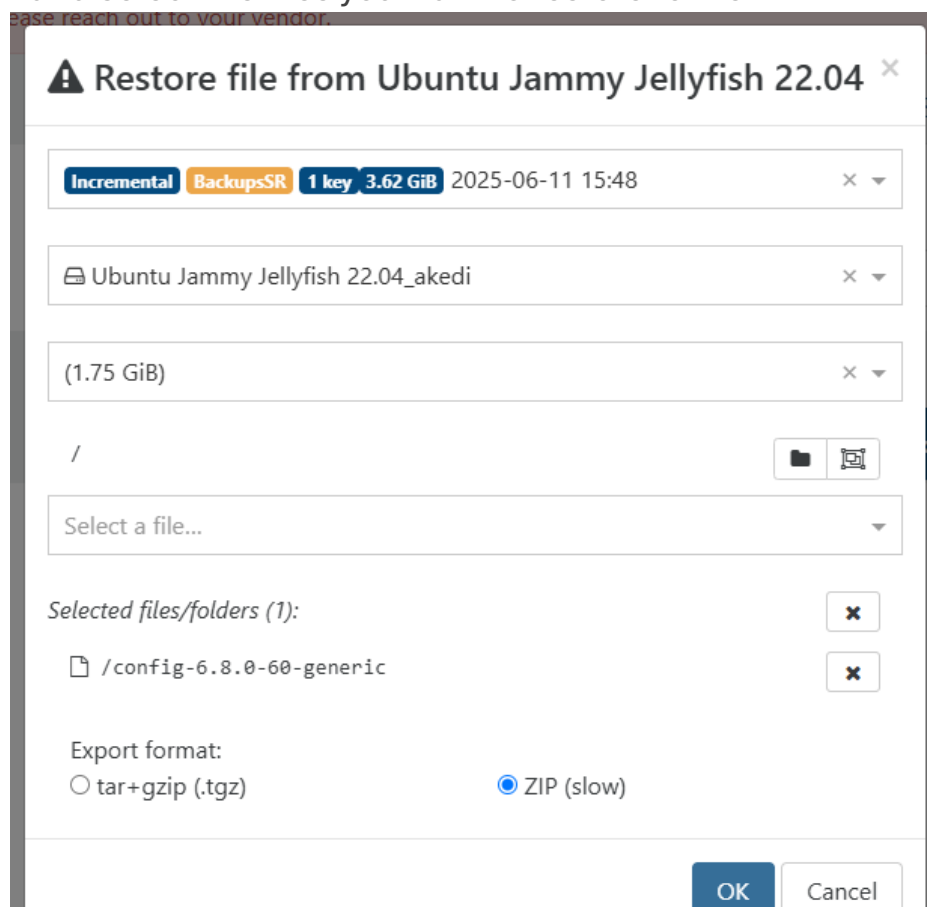


Figure 66: File restore selection wizard

6. Sequences Tab

Within the tab sequences this will show your all your available sequences that have been created and their status. Sequences are a set list of ordered steps that the Xcp-ng server will do in accordance with the configuration.

The screenshot shows the 'Sequences' tab in the Xcp-ng interface. At the top, there's a navigation bar with 'Backup' selected, and tabs for 'Overview', 'Sequences', '+ New', 'Restore', 'File restore', and 'Health'. Below this, the 'Sequences' section has a '+ New' button and a search bar. A table lists 3 items:

ID	Name	Sequence	Schedule
885f	Sequence 2	1. Unknown schedule 2. Unknown schedule	0 0 * * *
a9b0	Sequence Beta	1. <i>Unnamed schedule</i> (XO Configs) 2. <i>Rarely</i> (Metadata) 3. <i>Unnamed schedule</i> (Unknown job)	0 0 */14 */6 *
e95a	Sequence Alpha	1. <i>Unnamed schedule</i> (Regular backup) 2. <i>Unnamed schedule</i> (VM Replication) 3. <i>Unnamed schedule</i> (VM Mirror)	0 0 */7 */3 *

Figure 67: Sequences main tab

In the top right hand side, you can create another new sequence which will greet you with the following options for configuration. Here you can specify the steps that you want to take place. You can select backup jobs and other sequences in the list of steps; this will create a conjoined job that will process the first job on the list till the end and will even have its own independent schedule.

As shown in the image below we had added a sequence that will run the regular backup for hour ubuntu server after the ubuntu VM replication job and then mirror job in one swift operation. Also we have setup the schedule to run the job every 3 months on ever week.

Sequence

Name

1. Unnamed schedule (Regular backup)

2. Unnamed schedule (VM Replication)

3. Unnamed schedule (VM Mirror)

+ Add

Schedule sequence

ⓘ If your country participates in DST, it is advised that you avoid scheduling jobs at the time of change. e.g. 2AM to 3AM for US.

Month

January	February	March
April	May	June
July	August	September
October	November	December

3

Every month

Day

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

Figure 68: Sequences Recursion

Then lastly under the sequences tab after creation you have the option to either run sequence, edit it, or delete it.

Conclusions

Tristan's Conclusion

As of this projects beginning, I was assigned to make all the tasks on the XCP-ng server and from my experiences I have really enjoyed working with XCP-ng overall.

My dislikes about XCP-ng have been its clunky and outdated look or feel, and certain aspects were not very well documented and explained thus causing a lot of confusion. XOA the Xen Orchestra VM at times was rather slow and took longer than usual to perform basic functions.

On the other hand, as a hypervisor goes, it was very intuitive and easy to pick up, with some common knowledge I was about to operate most aspects of the Hypervisor relatively easily and swiftly. As one of the only two hypervisors I have operated so far in my experiences, this had been a pleasant experience overall and would not mind using this in the future.

Refik's Conclusion

I was using Proxmox for the tasks and in short, I liked the hypervisor operator. With most of the tasks I had no problems, and it was very simple, usually not too long or difficult to get what I wanted out of it.

I did not like how often the Proxmox would break, and I would have to reinstall the entire Hypervisor. It also gave me a lot of errors that after a bit of confusion resolved themselves. However, I did like the UI and the snappyness of the hypervisor.

Setup Guide Table

This section outlines the configurations for the servers that have been given with their respective Ips and installation locations and passwords for any reference now or in the future.

Tristan		Refik	
PROXMOX IP	10.0.10.31	PROXMOX IP	10.0.10.82
Prox Boot Drive	Kingston Voyager	Prox Boot Drive	Samsung
Prox Storage Drive	NAS	Prox Storage Drive	Toshiba
XCP IP	10.0.10.32	XCP IP	10.0.10.81
XCP Boot Drive	Samsung	XCP Boot Drive	Corsair Voyager
XCP Storage Drive	Western Digital	XCP Storage Drive	Western Digital
XCP VM IP	10.0.10.34	XCP VM IP	10.0.10.84
XOA WEB UI	ADMIN@ADMIN.NET ADMIN	XOA WEB UI	ADMIN@ADMIN.NET ADMIN
XCP VM Usr/Pwd	Xoa/test..123	XCP VM Usr/Pwd	Xoa/test..123
XCP VM Usr/Pwd	deltr691@school.lu/ test..123	XCP VM Usr/Pwd	mahre667@school.lu / test..123
XCP Windows Server	test..123	XCP Windows Server	test..123
XCP Ubuntu	Tristan/ 12341234	XCP Ubuntu	Refik/ 12341234

Figure 69: Setup information Table

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